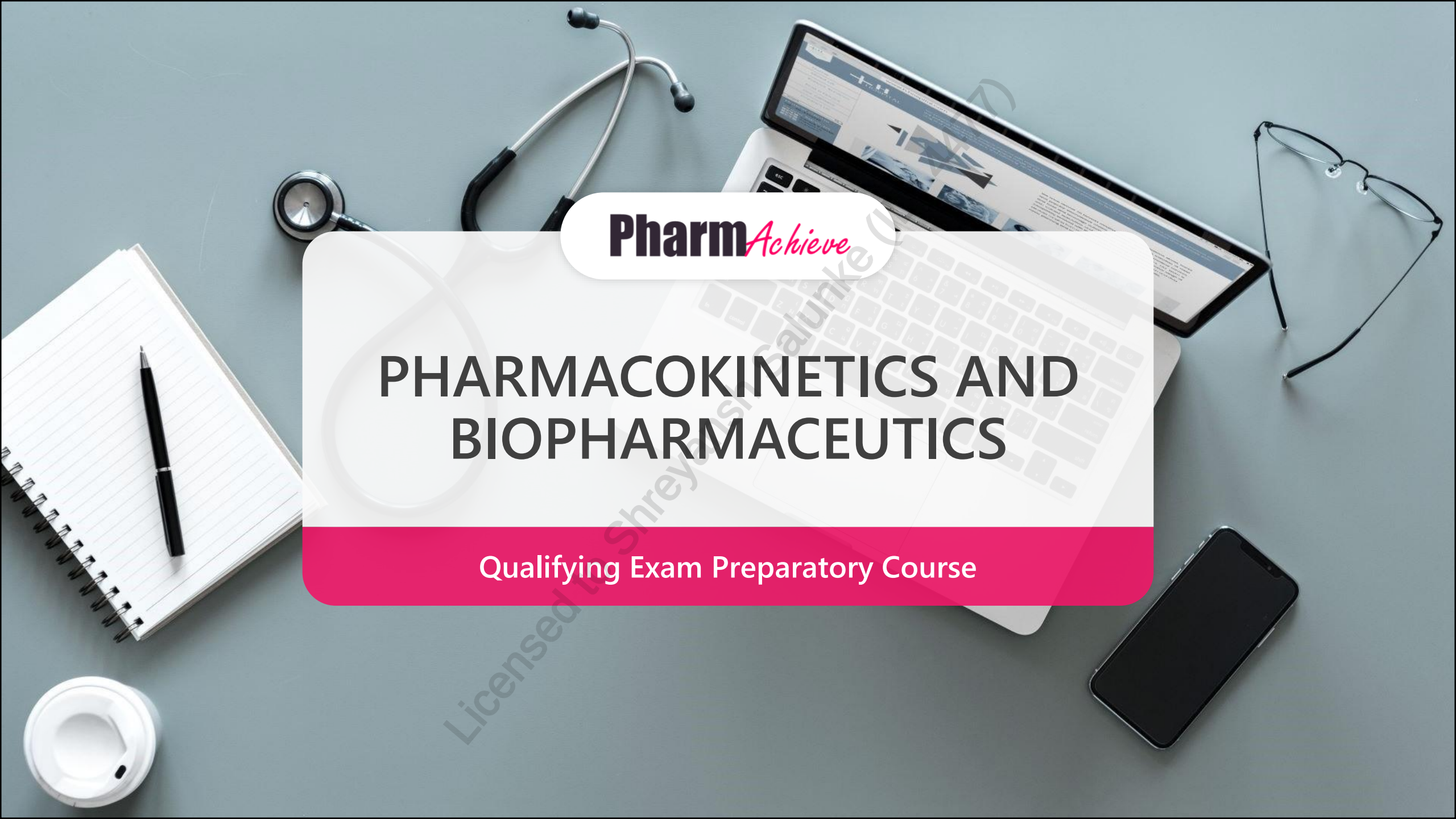




Pharm*Achieve*

PHARMACOKINETICS AND BIOPHARMACEUTICS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

AAG	Alpha-1-Acidic Glycoprotein	MATE	Multidrug and Toxic Compound Extrusion
ADME	Absorption, Distribution, Metabolism, Excretion	MRP	Multidrug Resistance Protein
ATP	Adenosine Triphosphate	OAT	Organic Anion Transporter
AUC	Area Under The Curve	OATP	Organic Anion Transporting Polypeptide
BCRP	Breast Cancer Resistance Protein	OCT	Organic Cation Transporter
BSEP	Bile Salt Export Pump	PEPT	Peptide Transporter
Ch_u	Hepatocyte Unbound Drug Concentration	P-gp	P-Glycoprotein
CL	Clearance	PK	Pharmacokinetics
C_p	Plasma Concentration	P_m	Permeability Of The Drug Through The Membrane
D	Dose	PO	Oral (per os)
F	Bioavailability	SA_m	Surface Area Of The Membrane
IV	Intravenous	V_d	Volume of Distribution
K_m	Michaelis-Menten Constant	τ	Dosing Interval

TABLE OF CONTENTS

Lecture	Topics
Pharmacokinetics Part 1	• Introduction to PK • Absorption • Distribution • Elimination • Metabolism • Compartment Models • IV bolus Drugs • Oral Drug Absorption
Pharmacokinetics Part 2	• Multiple Doses and Steady-State • Therapeutic Drug Monitoring • Non-linear PK • Interactions • Pharmacodynamics

EXPONENTS AND LOGARITHMS

Form = X^a

X is the base

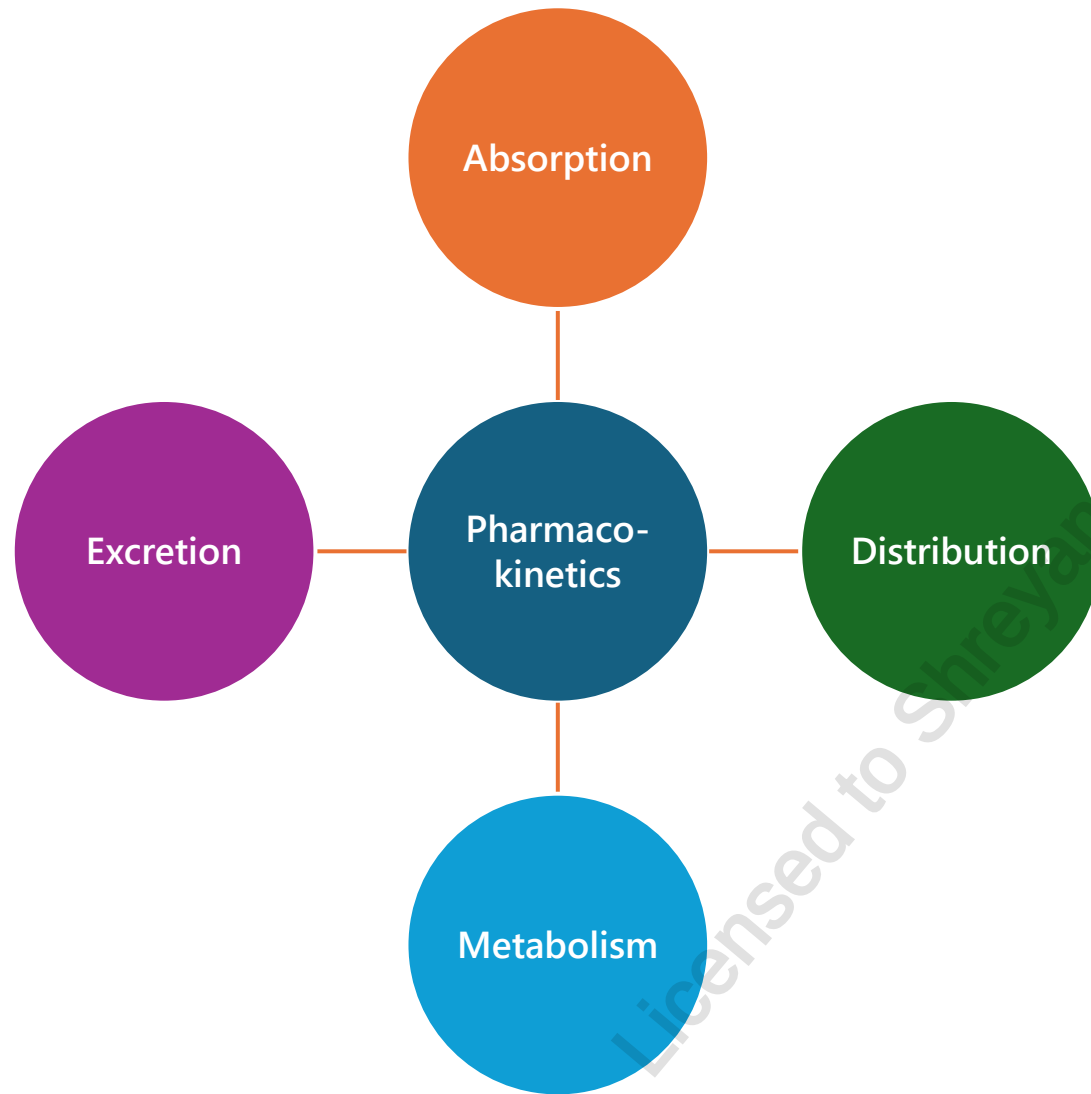
A is the exponent

If $X^b = Y$, then $\log_X Y = b$

- $X^0 = 1, X^1 = X$
- $X^{-a} = \frac{1}{X^a}$
- $X^a * X^b = X^{a+b}$
- $\frac{X^a}{X^b} = X^{a-b}$
- $X^{a^b} = X^{a*b}$
- $(X \cdot Y)^a = X^a \cdot Y^a$

- $\log_a a^x = x$
- $\log(XY) = \log X + \log Y$
- $\log \frac{X}{Y} = \log X - \log Y$
- $\log(X^y) = y \log X$
- $\log_a X = \frac{\log_b X}{\log_b a}$
- $\log(X + Y) \neq \log X + \log Y$

INTRODUCTION TO PHARMACOKINETICS



Importance of studying ADME:

- Identify drug/patient factors that determine the rate of each process
- Identify a way to quantify each process using a single parameter
- Derive a mathematical expression for the relationship between a drug's plasma concentration and time

ABSORPTION

- **Transport of drug from the site of administration to systemic circulation**
- Affected by various drug and patient-specific factors:
 - Drug-specific: solubility, pH, particle size, dissolution rate, route of administration
 - Patient-specific: age, blood flow at absorption site, GI content
- Drug must cross the cell membrane, regardless of absorption site!

Passive Diffusion	Facilitated Diffusion	Active Transport
• Net movement of drug from an area of high → low concentration • Most common	• Carrier- or channel-mediated transport • Does not require energy • Cannot transport against concentration gradient	• Carrier- or channel-mediated transport • Requires energy/ATP • Enables movement from low → high drug concentrations

FICK'S LAW

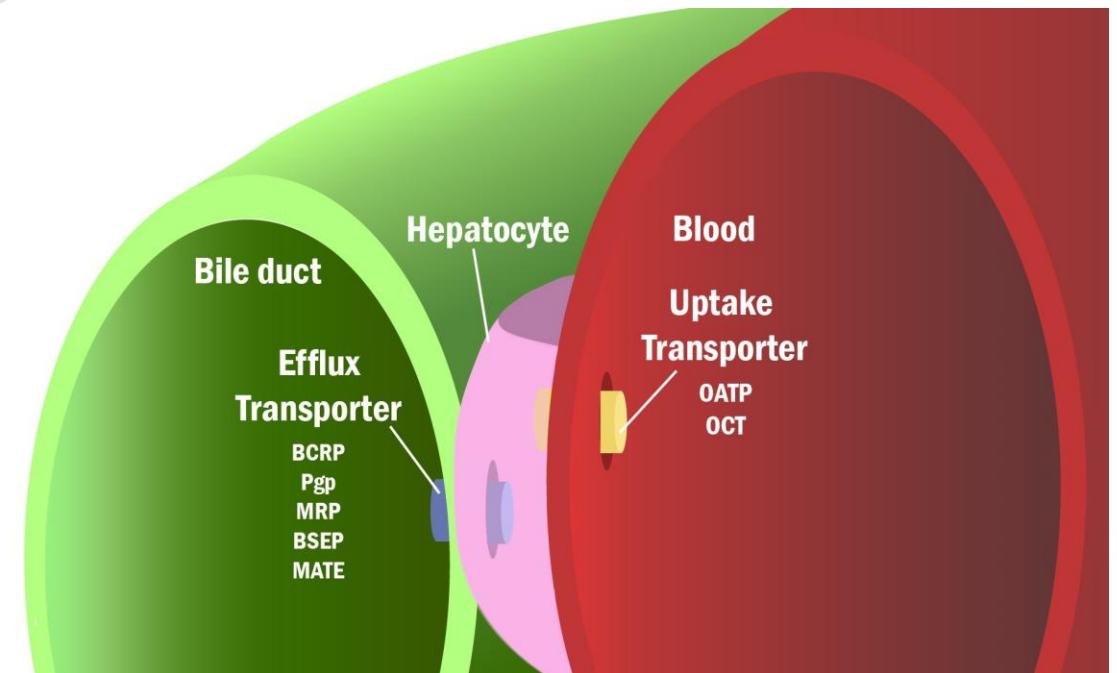
Diffusion is governed by Fick's Law:

$$\frac{dA_b}{dt} = P_m \cdot SA_m \cdot (C_{u1} - C_{u2})$$

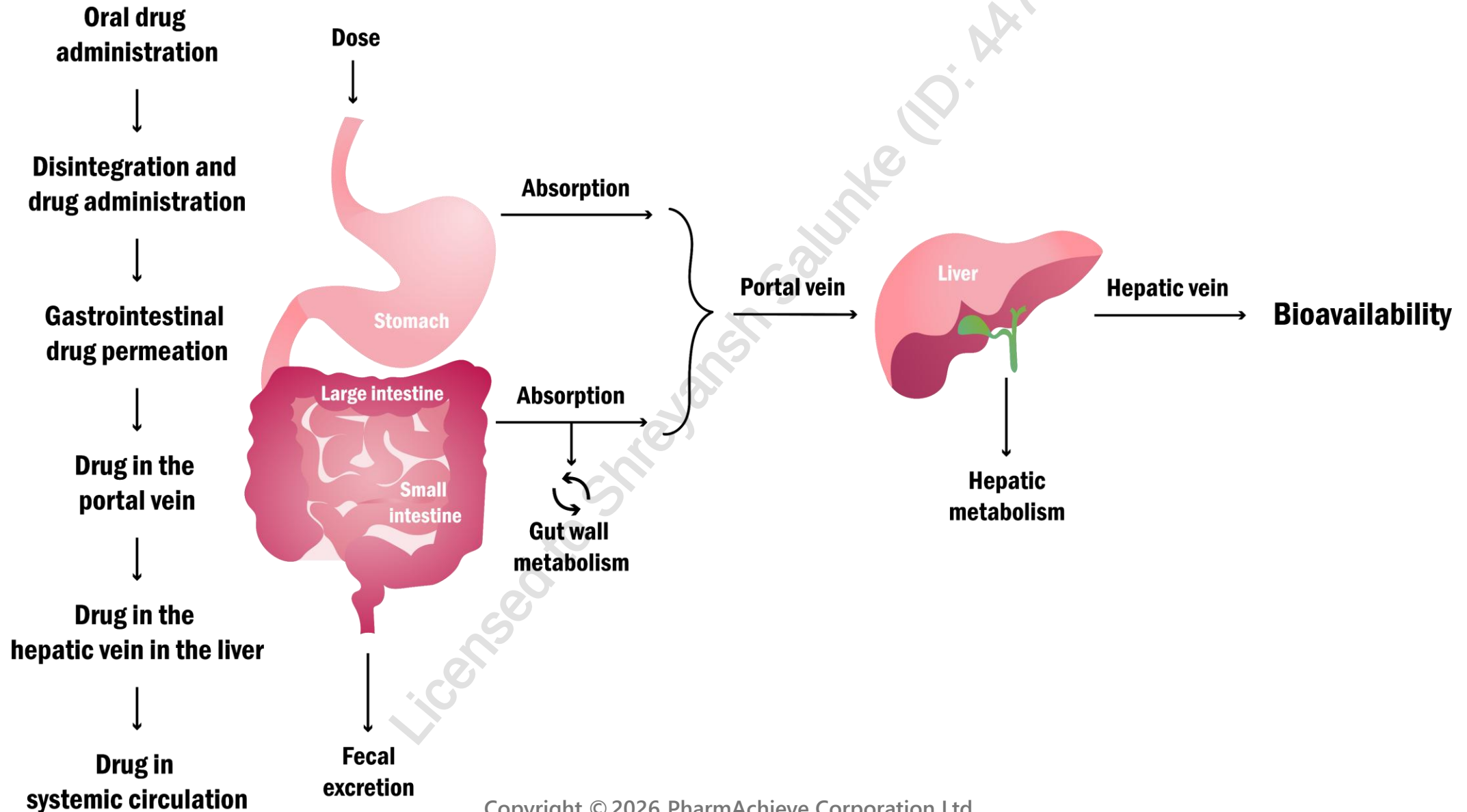
- dA_b/dt is rate of drug diffusion per unit time (mg/h)
- P_m is the permeability of the drug through the membrane (cm/h)
- SA_m is the surface area of the membrane (cm²)
- C_{u1} is the higher unbound drug concentration (mg/mL)
- C_{u2} is the lower unbound drug concentration (mg/mL)

TRANSPORTERS

- **Membrane-bound proteins that facilitate movement of molecules in and out of the cell**
- **Influx transporter:** transport drugs **into** the cell
 - Bidirectional facilitated diffusion and do not require ATP
 - Vital influx transporters: OATP, OAT, OCT, PEPT
- **Efflux transporter:** transport drugs **out** of the cell
 - Require ATP to function
 - Vital efflux transporters: P-gp, MRP, BCRP, BSEP, MATE



OVERVIEW OF ORAL DRUG ABSORPTION



BIOAVAILABILITY

- **Bioavailability factor (F):** fraction of the dose that reaches systemic circulation and ranges between 0 and 1
 - Three determinants of F: physiochemical characteristics, biological factors, drug formulation
- **Three types of bioavailability**
 1. **Fraction absorbed (F_a):** the fraction that is absorbed intact in the intestinal membrane
 2. **Intestinal bioavailability (F_g):** the fraction of the drug in the enterocytes that escapes metabolism
 3. **Hepatic bioavailability (F_h):** the fraction that enters the liver and escapes first-pass metabolism

DISSOLUTION

- **The process in which the oral dose dissolves into the GI fluid**
 - Determined by: hydrophilicity, lipophilicity, crystalline form of the drug, particle size
- **pH-partition hypothesis:** a nonionized species will more readily partition into a lipophilic solvent than an ionized species
 - Whether a drug is charged depends on pKa and pH of its environment
- **Drugs are formulated as salts to improve wetting**
 - Leads to faster dissolution and increased rate of absorption
 - E.g. fluoxetine HCl, naproxen sodium, etc.

RATE OF DISSOLUTION

The Noyes-Whitney equation describes the rate of dissolution of a drug

$$\frac{dC}{dt} = \frac{D \cdot S}{h} \cdot (C_s - C)$$

- C is the concentration in the gastrointestinal fluid
- t is time
- S is surface of the drug
- D is the diffusion coefficient
- h is thickness of the diffusion layer surrounding the solid
- C_s is the solubility of the drug

RATE OF DRUG ABSORPTION

- Rate of drug absorption is controlled by slowest step
- When dissolution rate controls absorption, the [Noyes-Whitney equation](#) describes the rate
- The drug concentration in gastrointestinal fluid will be much smaller than the drug solubility
- The process then approximates a first-order process

Concentration is much smaller than solubility ($C \ll C_s$)

$$\frac{dC}{dt} = \frac{D \cdot S}{h} \cdot (C_s)$$

- t is time
- S is surface of the drug
- H is thickness of the diffusion layer surrounding the solid
- D is the diffusion coefficient
- C_s is the solubility of the drug

RATE OF DRUG ABSORPTION

- When **membrane penetration controls drug absorption**, the rate is approximated by Fick's law
- Drug is rapidly diluted into a large volume and process approximates a first-order process

Drugs that diffuse across the membrane are diluted into a large volume:

$$\frac{dA_b}{dt} = \text{Constant} \cdot (C_{u1})$$

- **Constant** is the product of P_m and SA_m
- The rate of diffusion is proportional to the concentration of the drug

RATE OF DRUG ABSORPTION

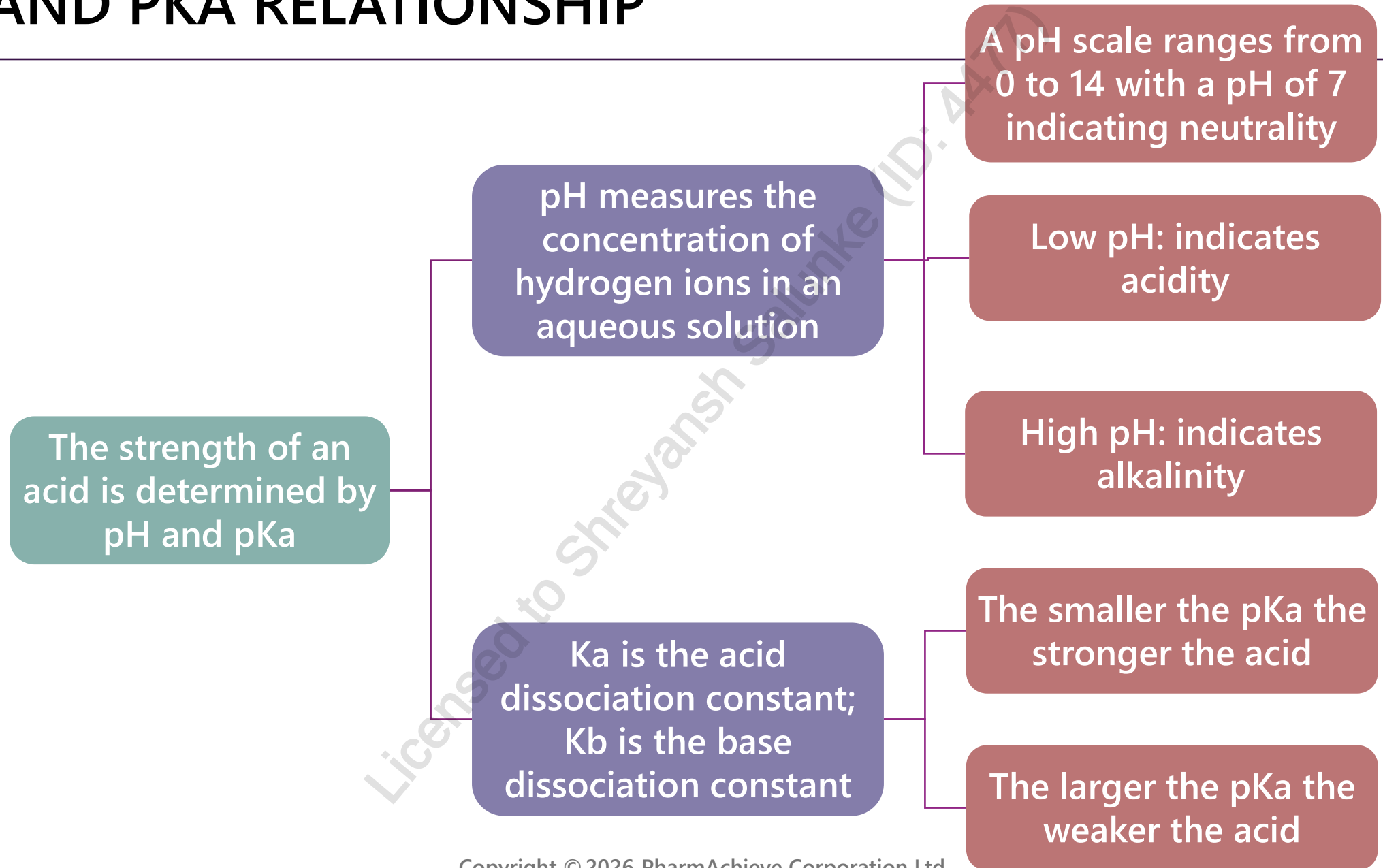
- Dissolution and membrane penetration approximate a first order process
- Orally administered drugs assumed to follow first-order kinetics

The rate of absorption is a first-order process:

$$\frac{dAb}{dt} = A_{GI} \cdot k_a$$

- **Ab** is the amount of drug in the body
- **t** is time
- **A_{GI}** is amount of drug in the GI tract
- **k_a** is the rate constant for absorption

PH AND PKA RELATIONSHIP



PERCENT IONIZATION

- A weak acid or base partially ionizes into its ions in an aqueous solution
- Example: Acetic acid (*weak acid*) and ammonia (*weak base*)
- Percent (%) ionization: Amount of acid or base that dissociates into its ions at a specific concentration
- Strong acids and bases fully dissociate into ions; their percent ionization is 100%

$$\% \text{ ionization for weak acid} = \frac{100}{1 + \text{antilog}(\text{pK}_a - \text{pH})}$$

$$\% \text{ ionization for weak base} = \frac{100}{1 + \text{antilog}(\text{pH} - \text{pK}_a)}$$

K_a : equilibrium constant of an acid
 pK_a scale is defined by: $\text{pK}_a = -\log K_a$

$$\text{Antilog } x = 10^x$$

PRACTICE QUESTION 1

1. If the pKa of acetylsalicylic acid is 3.5, what fraction of the drug will be ionized at a pH of 5.4?

PRACTICE QUESTION 1- ANSWER

1. If the pKa of acetylsalicylic acid is 3.5, what fraction of the drug will be ionized at a pH of 5.4?

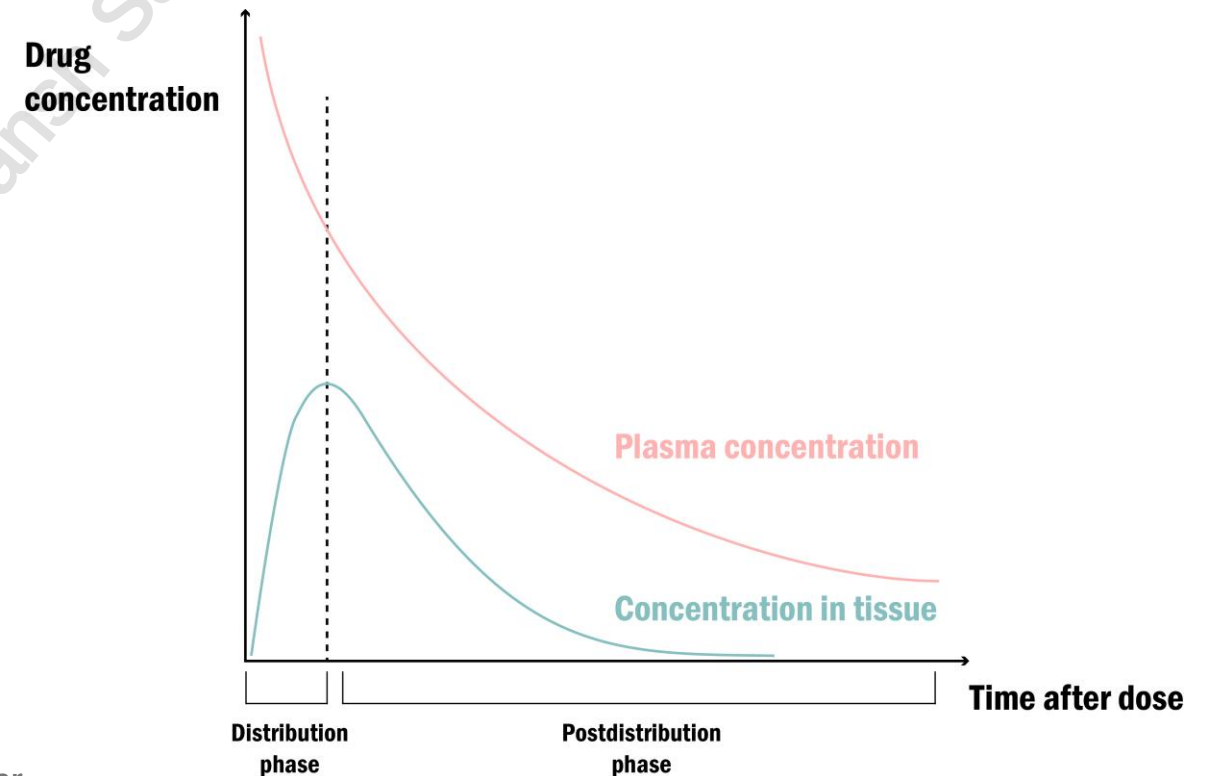
$$\% \text{ ionization} = \frac{100}{1 + \text{antilog}(pKa - pH)}$$

$$\% \text{ ionization} = \frac{100}{1 + \text{antilog}(3.5 - 5.4)}$$

$$\% \text{ ionization} = \mathbf{98.8\%}$$

DRUG DISTRIBUTION

- Describes how rapid and to what extent drug in plasma gets up taken by tissue
- Challenging to measure tissue concentration; **plasma concentration measured instead**
- The **effect only occurs when a drug reaches the site of action**
 - Only a fraction of overall dose reaches
- Drug distribution is **driven by passive diffusion of the unbound drug**
 - **Distribution phase:** after administration, the plasma drug concentration is much higher than that in the tissue → drives diffusion of drug into tissue
 - **Post-distribution phase:** equilibrium is established between tissue and plasma, ratio between concentrations rises and falls in parallel



VOLUME OF DISTRIBUTION

- **Volume of distribution (Vd)** is the primary distribution parameter for a drug
- Vd has units of volume, but is **not a physiological volume**
- **Vd reflects how a drug will distribute throughout the body**

Vd is constant for a drug and is the ratio of the amount of drug in the body and the plasma concentration at a specific time

$$Vd = \frac{Ab}{Cp}$$

- **Ab** is the **amount of drug in the body at a specific time**
- **Cp** is the **plasma concentration at the same time**

IMPLICATIONS OF APPARENT VOLUME OF DISTRIBUTION

	Low V_d	High V_d
Causes	• Large • Hydrophilic • Plasma protein bound • Impermeable membranes	• Small • Hydrophobic • Tissue protein-bound • Permeable membranes
Implications	• Low dose to saturate plasma • Fast clearance – if free drug • Tissues receiving equal exposure	• Allows targeting of tissues • Can cause toxic buildup • Concentrated in specific tissues

VOLUME OF DISTRIBUTION OF SELECTED DRUGS

FYI

Drug/Volume	Vd (L)
Plasma	3
Heparin	4.2
Warfarin	8
Extracellular fluid	15
Aminoglycosides	20
Total body water	40
Phenytoin	45
Diazepam	77

Drug/Volume	Vd (L)
Lidocaine	105
Morphine	245
Propranolol	420
Digoxin	500
Felodipine	700
Nortriptyline	1260
Amiodarone	4600
Chloroquine	14,000

PRACTICE QUESTION 2

2. Theophylline has a volume of distribution of 0.5 L/kg. A 70-kg adult has a plasma concentration of 12 mg/L. How much theophylline is in their body?

PRACTICE QUESTION 2- ANSWER

2. Theophylline has a volume of distribution of 0.5 L/kg. A 70-kg adult has a plasma concentration of 12 mg/L. How much theophylline is in their body?

$$Vd = \frac{Ab}{Cp}$$

$$Ab = Cp \cdot Vd$$

$$Ab = 12 \frac{mg}{L} \cdot 0.5 \frac{L}{kg} \cdot 70 kg$$

$$Ab = 420 mg$$

PRACTICE QUESTION 3

3. A patient is given 500 mg of drug X by IV injection. Their plasma concentration at time zero was measured and found to be 0.1 mg/L. What is the volume of distribution for drug X?

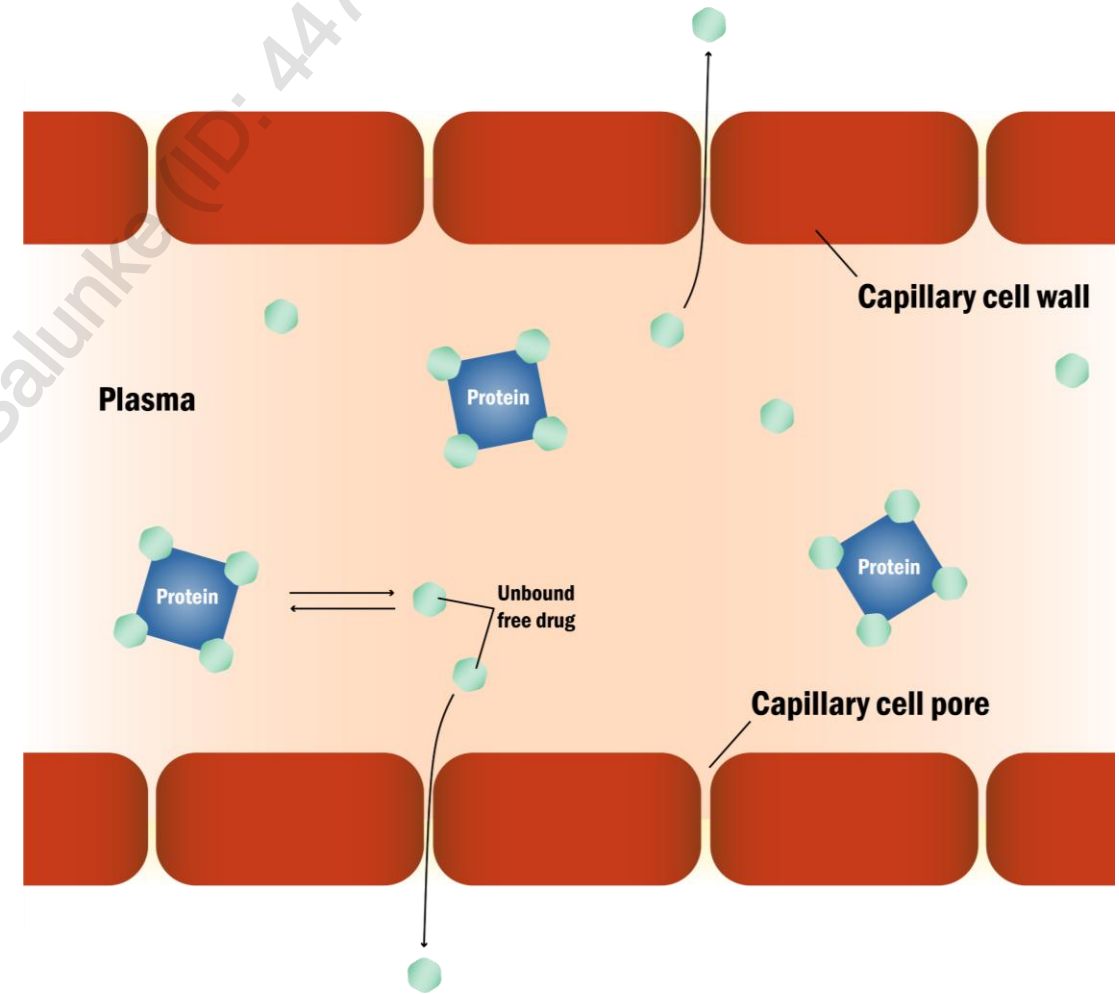
PRACTICE QUESTION 3- ANSWER

3. A patient is given 500 mg of drug X by IV injection. Their plasma concentration at time zero was measured and found to be 0.1 mg/L. What is the volume of distribution for drug X?

$$Vd = \frac{Ab}{Cp}$$
$$Vd = \frac{500\text{ mg}}{0.1 \frac{\text{mg}}{\text{L}}}$$
$$Vd = 5000\text{ L}$$

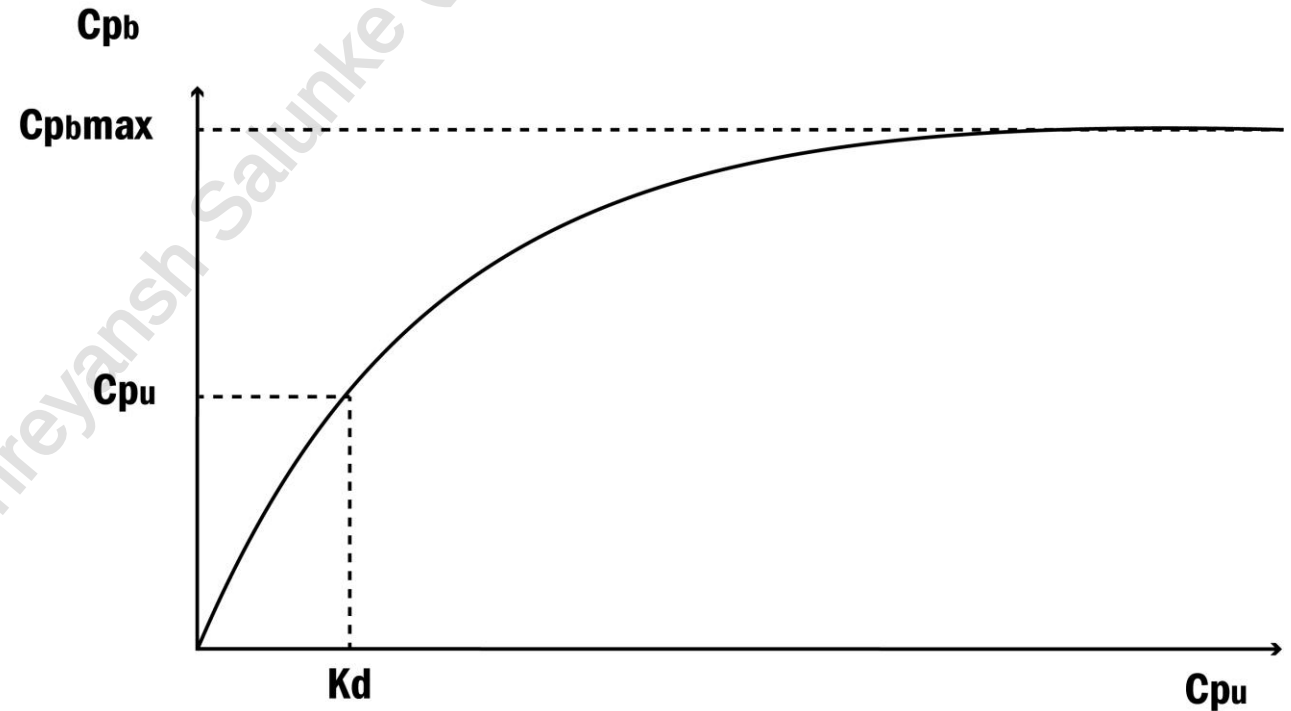
PROTEIN BINDING

- Drug **distribution can vary** due to plasma protein binding
- Drug binding to proteins/molecules is **reversible**
- Only **unbound drug** is available to participate in the concentration gradient
 - Hence, the extent of drug-protein binding will impact the quantity of drug able to diffuse
- **Albumin, AAG, and lipoproteins** are **primary plasma proteins** involved in drug binding



PROTEIN BINDING

- Drug-protein binding = **saturation** or “**capacity-limited**” process
- $C_{p(\text{unbound})} < K_d$ = drug-protein binding is directly proportional
- $C_{p(\text{unbound})} > K_d$ = amount of bound drug reaches upper limit, $C_{p(\text{bound})}$, max (**binding sites become saturated**)



LIVER METABOLISM

- Many tissues have enzymes for metabolism. The liver is the main site of metabolism.
- Primary purpose of hepatic metabolism is to **make the drug easier to excrete** (i.e., in urine and/or bile)

Metabolism occurs in 2 phases...

Phase I	Phase II
• Small chemical modifications of the drug molecule • Oxidation, reduction, hydrolysis	• Addition of functional group to yield polar metabolite • Usually results in drug inactivation but can make drug more active than parent compound • Conjugation reactions (e.g., glucuronidation)

DRUG METABOLISM

- Drug metabolism is an enzymatic process
- The process is initiated by the reversible binding of the substrate to the enzyme
- The enzyme converts the drug to its metabolite, and they separate

Drug metabolism process is described as follows



- **D** is the unbound drug concentration
- **E** is the enzyme concentration
- **k₁** is the forward rate constant for the drug enzyme interaction
- **k₂** is the backwards rate constant for the drug enzyme interaction
- **DE** is the drug-enzyme complex concentration
- **k_{cat}** is the rate constant for metabolite formation
- **M** is the metabolite concentration

SPEED OF METABOLISM

Metabolism is a capacity limited process

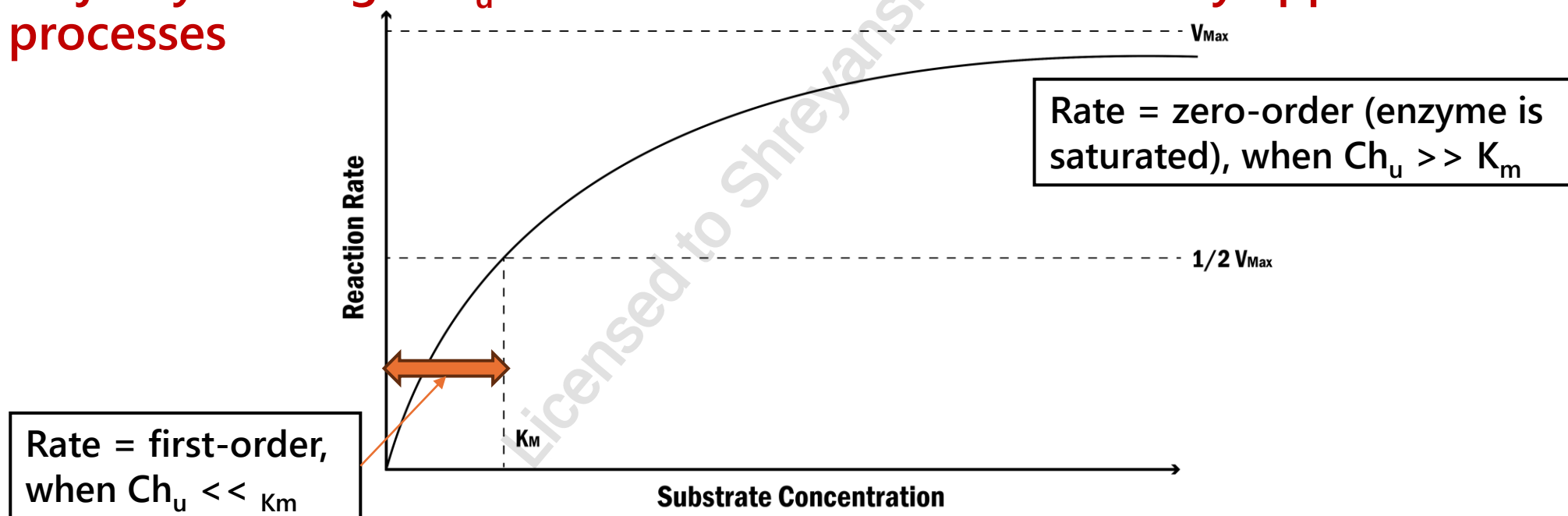
The Michaelis-Menten equation describes the velocity of metabolism

$$V = \frac{V_{max} \cdot Ch_u}{K_m + Ch_u}$$

- **V** is the **velocity**
- **V_{max}** is the **maximum metabolism speed achieved at enzyme saturation**
- **Ch_u** is the **hepatocyte unbound drug concentration**
- **K_m** is the **Michaelis-Menten constant**

SPEED OF METABOLISM

- When **velocity** is **exactly half** Ch_u is equal to K_m
- When $Ch_u \gg K_m$, the enzyme is saturated, and the **rate** is a **zero-order process**
- When $Ch_u \ll K_m$, the **rate** **approximates a first-order process**
- Overall, rate of metabolism is mixed order and nonlinear
- **Majority of drugs Ch_u is much lower than K_m and they approximate first-order processes**



ELIMINATION

- Elimination incorporates all processes involved in removal of a parent drug from the body
- **Renal excretion:** Parent drug is excreted out of the body via the urine
- **Liver metabolism:** Parent drug is converted into metabolites by the liver
 - Once the drug is converted to its metabolites, it is considered eliminated
- Both the rate of excretion and the rate of metabolism are considered first order processes

CLEARANCE

- **Clearance:** The efficiency in which the liver/kidney can extract drug from plasma and eliminate it
 - **Fundamental PK parameter for elimination** with units of volume/time
 - Total clearance = renal clearance + hepatic clearance
 - **Constant**; altered by disease, medication, changes in organ function
- Large clearance + small volume of distribution = rapid elimination + short half-life
- Small clearance + large volume of distribution = slow elimination + long half-life

Clearance is the proportionality between the rate of elimination and the plasma concentration

$$\text{rate of elimination} = Cl \cdot C_p$$

- **Cl** is the **clearance**
- **C_p** is the **plasma concentration**

CLEARANCE SUMMARY

1. Clearance is the primary parameter for elimination and expresses the ability of the kidneys and liver to remove drug from the systemic circulation
2. Clearance is expressed in terms of the equivalent volume of blood or plasma completely cleared of drug per unit time as it passes through the organ
3. Clearance is dependent on the blood flow to the organ (Q) and on the ability of the organ to extract the drug from the bloodstream and eliminate it (E)
4. Clearance is the constant of proportionality between the rate of drug elimination and the plasma concentration
5. Clearances are additive, and total body clearance is often expressed as the sum of its two main component clearances: renal and hepatic clearances.

ELIMINATION RATE CONSTANT

- The rate of elimination is a first order process

The elimination rate constant is the ratio between the clearance and the volume of distribution

$$k = \frac{Cl}{Vd}$$

- **k** is the elimination rate constant
- **Cl** is the drug clearance
- **Vd** is the volume of distribution

ELIMINATION HALF-LIFE

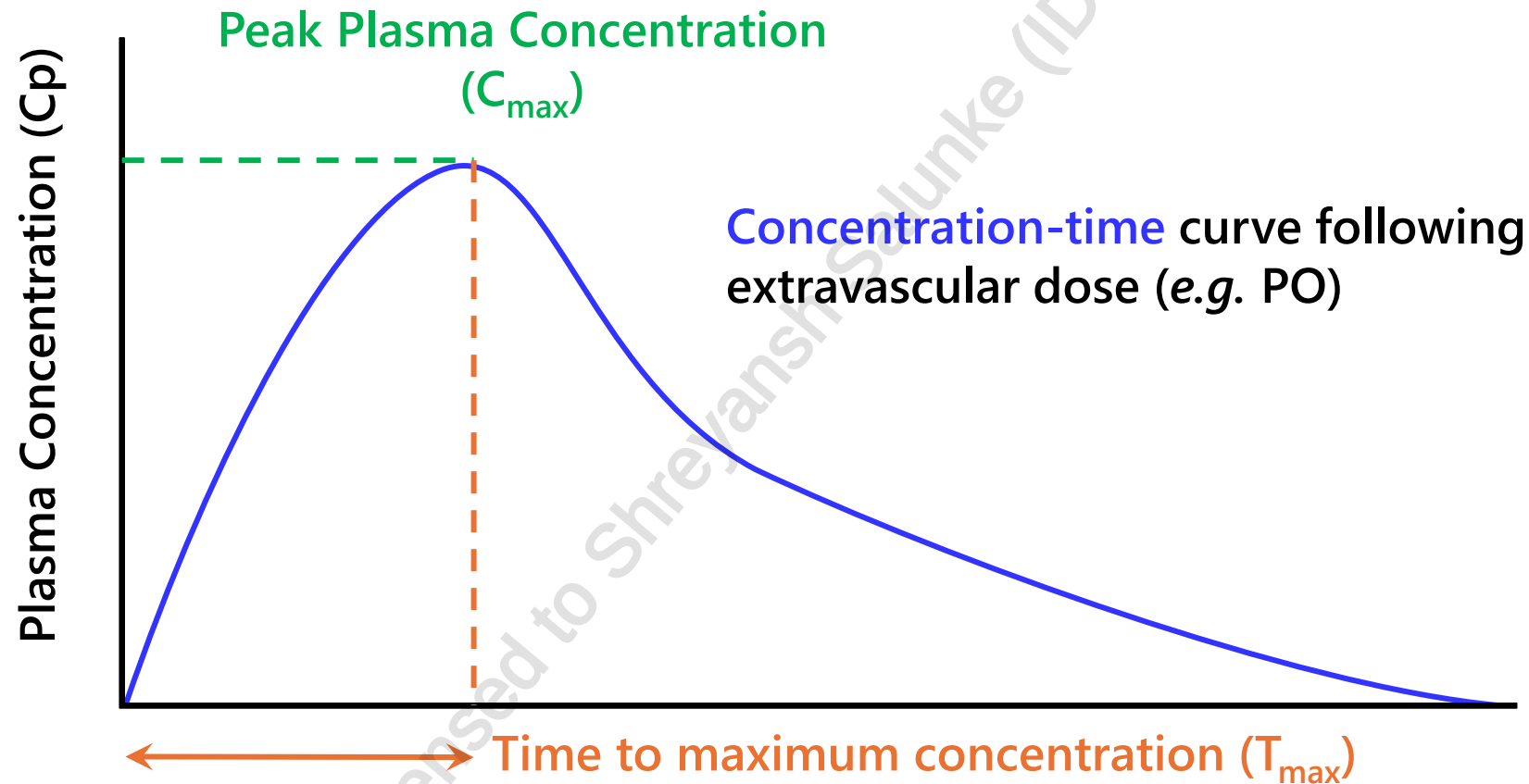
- Half-life and elimination measure the speed of drug elimination
- Reciprocal forms of each other
- Large elimination rate constant = short half life
- Low elimination rate constant = long half-life

The elimination half life is:

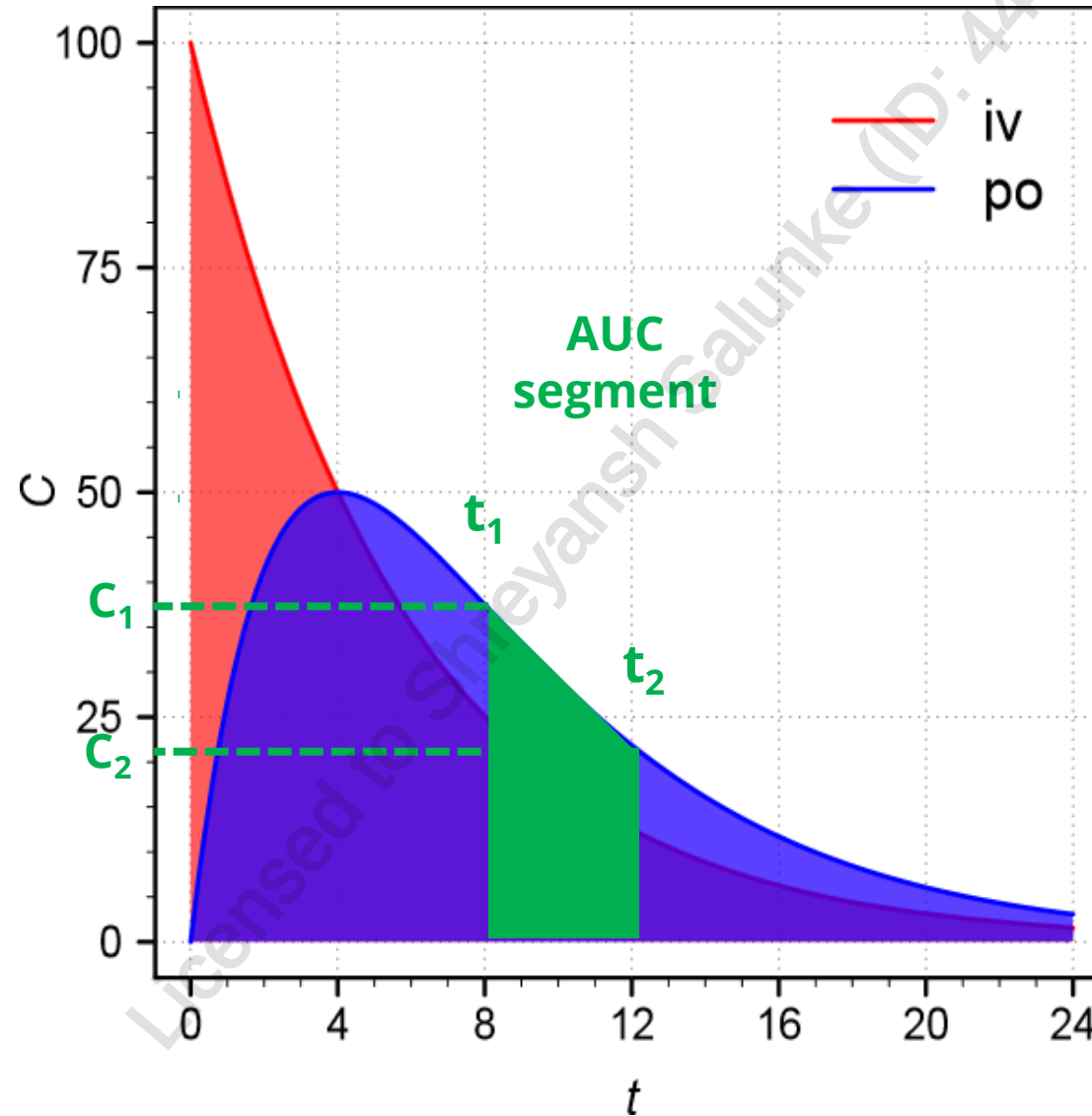
$$t_{1/2} = \frac{0.693}{k} = 0.693 \frac{Vd}{Cl}$$

- **k** is the elimination rate constant
- **Cl** is the drug clearance
- **Vd** is the volume of distribution

CONCENTRATION-TIME CURVE



AREA UNDER THE CURVE (AUC)



CLEARANCE USING AUC

The effective dose is eliminated by infinity and we can derive a general formula for the clearance of oral medications

The clearance can be expressed using the area under the curve:

$$Cl = \frac{D_{iv}}{AUC_0^\infty} = \frac{F \cdot D}{AUC_0^\infty}$$

- **Cl** is the drug clearance
- **D_{iv}** is the IV dosage
- **F** is the bioavailability
- **D** is the oral dose
- **AUC₀[∞]** is the area under the curve from zero to infinity

PRACTICE QUESTION 4

4. A new drug is currently being investigated and it is found to have a volume of distribution of 250 L and a clearance of 1.5 L/h. What is the elimination half-life of this new drug?

PRACTICE QUESTION 4- ANSWER

4. A new drug is currently being investigated and it is found to have a volume of distribution of 250 L and a clearance of 1.5 L/h. What is the elimination half-life of this new drug?

$$k = \frac{Cl}{Vd}$$

$$k = \frac{1.5 \frac{L}{h}}{250 L}$$

$$k = \frac{0.006}{h}$$

$$t_{1/2} = \frac{0.693}{k}$$

$$t_{1/2} = \frac{0.693}{0.006 h^{-1}} = 115.5 h$$

RENAL ELIMINATION

Filtration:

- Blood passes through the glomerular capillaries and is filtered into Bowman's capsule
- Filtrate does not contain proteins (they are too large to pass through)

Reabsorption:

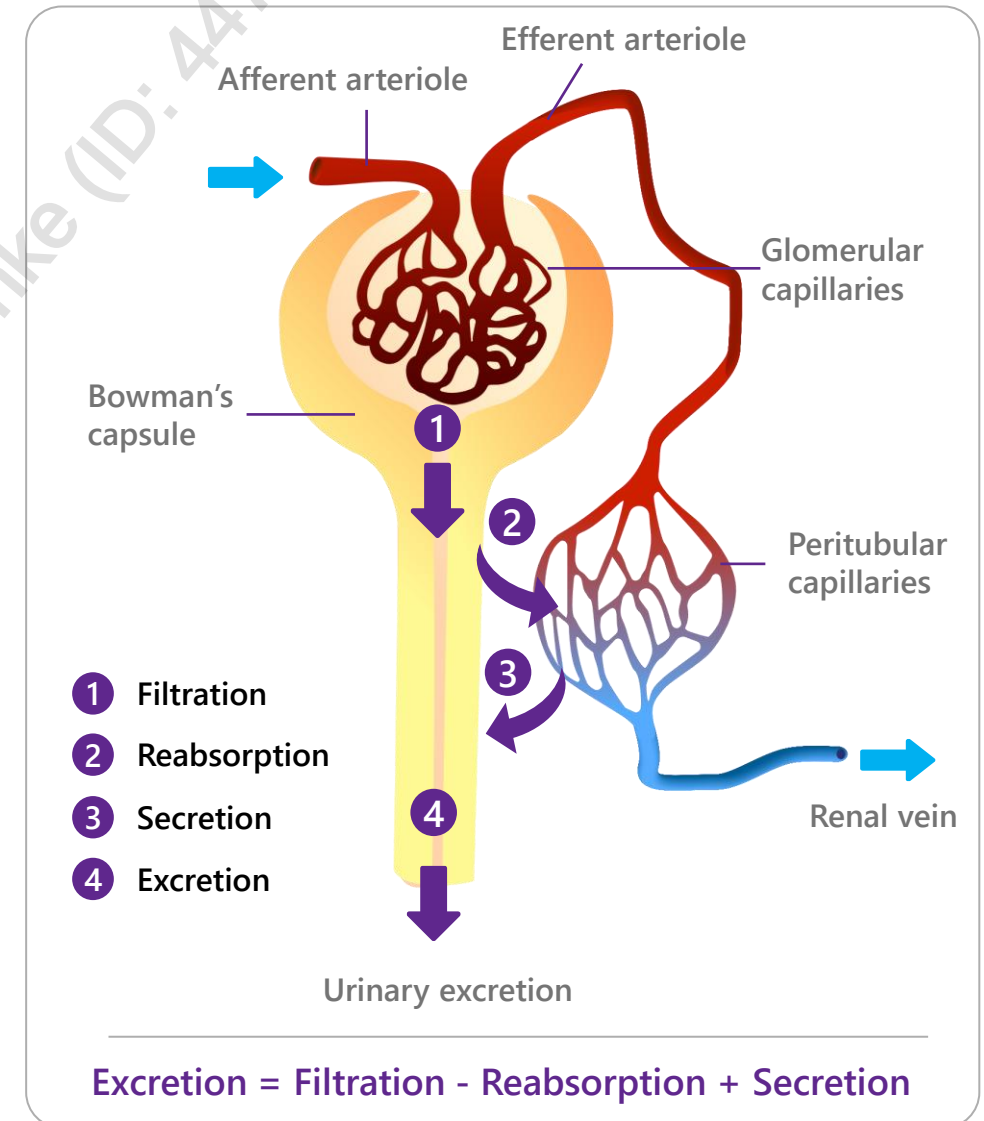
- Important nutrients are reabsorbed into the peritubular capillaries, from the renal tubules and collecting ducts
- Driven by concentration gradient
- pH can determine ionization

Secretion:

- Waste products are further secreted into the filtrate
- Primarily excretes weak acids
- Can reduce bioavailability

Excretion:

- Substances that are not reabsorbed become urine and are excreted from the collecting duct



RENAL CLEARANCE

- Renal System is a major site of drug elimination
- $CL_{\text{Renal}} = \frac{\text{Filtration} + \text{Secretion} - \text{Absorption}}{C_p}$
- Average clearance = 120 mL/min
- Target for changing drug bioavailability and $\frac{1}{2}$ life
- Renal dysfunction can result in abnormally high or even toxic C_p
 - Can be caused by vasoconstriction, blockage, infection, kidney stones
 - Glomerular failure can lead to excessive filtration and drug depletion
 - Restricted filtration can lead to reduced elimination
 - k_{el} can be recalculated by monitoring C_p to readjust dosage calculations

EXCRETION RATE AND PLASMA CONCENTRATION

- There are several ways to determine drug renal clearance
- The easiest method is to periodically collect urine samples and to create a rate of excretion vs plasma concentration plot
- This plot is linear, and the slope is equal to the renal clearance

Renal clearance is the relationship between excretion rate and plasma concentration

$$\frac{dAu}{dt} = Cl_r \cdot C_p$$

- Cl_r is the total clearance rate
- C_p is the plasma concentration
- Au is the amount of drug in the urine

PRACTICE QUESTION 5

5. The data is as follows:

Collection period: 0 – 24 h

Volume of urine collected: 1400 mL

Urinary creatinine concentration: 1.14 mg/mL

Serum creatinine concentration: 1.15 mg/dL

What is the patient's creatinine clearance?

PRACTICE QUESTION 5- ANSWER

5. The data is as follows:

Collection period: 0 – 24 h

Volume of urine collected: 1400 mL

Urinary creatinine concentration: 1.14 mg/mL

Serum creatinine concentration: 1.15 mg/dL

What is the patient's creatinine clearance in mL/min?

$$\text{amount of creatinine excreted in urine} = 1.14 \frac{\text{mg}}{\text{mL}} \cdot 1400 \text{ mL} = 1596 \text{ mg}$$

$$\text{average rate of drug clearance} = \frac{1596 \text{ mg}}{24 \text{ hours}} = 66.5 \frac{\text{mg}}{\text{h}}$$

$$\text{creatinine clearance} = \frac{66.5 \frac{\text{mg}}{\text{h}}}{11.5 \frac{\text{mg}}{\text{L}}} = 5.78 \frac{\text{L}}{\text{h}} \cdot \frac{\text{h}}{60 \text{ mins}} \cdot \frac{1000 \text{ mL}}{\text{L}} = 96.33 \frac{\text{mL}}{\text{min}}$$

HEPATIC CLEARANCE

- Liver metabolizes drugs – deactivating them and enabling excretion
- **First pass effect:** removal of drug by GI tract and liver prior to entering systemic circulation
- Drugs can inhibit/induce hepatic activity → can alter bioavailability of other drugs
- Liver failure can result in abnormally high C_p
 - Hepatic failure is common in cases of cirrhosis/toxic drugs
 - Can lead to decreased biotransformation and elimination
 - Can drastically reduce C_p of prodrugs that rely on hepatic transformation
 - k_{el} can be recalculated by monitoring C_p to readjust dosage calculations

HEPATIC CLEARANCE

- **Hepatic clearance:** the volume of fluid completely cleared of drug by the liter per unit time
- Maximum hepatic clearance = hepatic blood flow
- **Extraction ratio:** function of the ability for the drug to get from the blood into the hepatocyte and the ability of the hepatocyte enzymes to metabolize the drug
- Ability of hepatic enzyme to metabolize the drug is the drugs intrinsic clearance (Cl_{int})

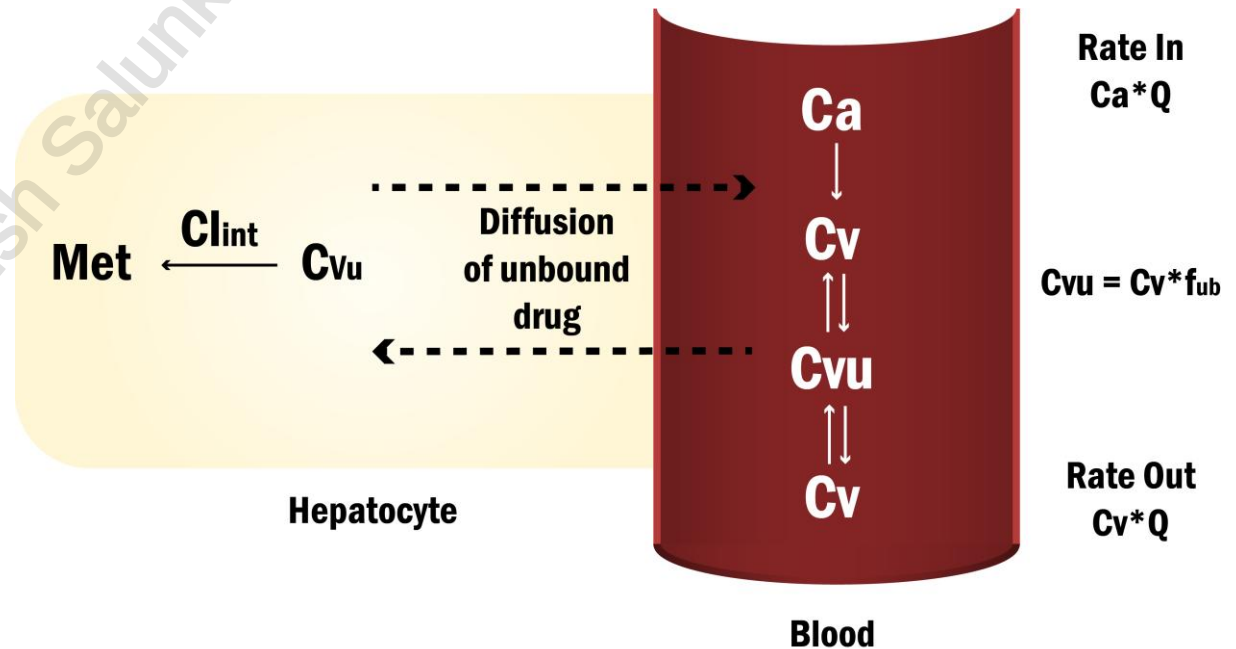
Hepatic clearance is a function of the blood flow and the extraction ratio

$$Cl_{bh} = Q \cdot E$$

- Cl_{bh} is the hepatic blood clearance
- Q is the hepatic blood flow
- E is the hepatic extraction ratio

WELL-STIRRED MODEL

- Hepatic clearance is based on the Well-Stirred model where **drug molecules are assumed to be distributed homogenously**
- Several assumptions are made:
 1. Only unbound drug in the blood is subject to elimination
 2. No membrane transport barrier
 3. No concentration gradient of the drug within the liver
 4. The concentration of the drug within the liver is equal to that in the venous blood
 5. Linear kinetics



WELL-STIRRED MODEL

The rate of metabolism is equal to the difference between drug presentation to the liver and the rate of drug exit from the liver

$$Cl_h = \frac{Q \cdot fu \cdot Cl_{int}}{Q + fu \cdot Cl_{int}}$$

- Cl_h is the hepatic clearance
- Q is the hepatic blood flow
- fu is the fraction of drug unbound in the blood
- Cl_{int} is the intrinsic clearance

Hepatic extraction ratio $E_H = \frac{fu \times Cl_{int}}{Q_H + fu \times Cl_{int}}$

HIGH VS LOW HEPATIC EXTRACTION RATIO

HIGH HEPATIC EXTRACTION RATIO

$$(Cl_{b_h} \approx Q)$$

High extraction ratio ($E > 0.7$) – drug is extensively cleared from blood by liver

Clearance is dependent primarily on **hepatic blood flow**

High extraction ratio drugs undergo increased first-pass extraction and have poor bioavailability (e.g. morphine, lidocaine, verapamil, nitroglycerin)

LOW HEPATIC EXTRACTION RATIO

$$(Cl_{b_h} \approx f_{u_b} \cdot Cl_{int})$$

Low extraction ratio ($E < 0.3$) – drug is not efficiently cleared from blood by liver

Clearance is dependent on **amount of free drug and intrinsic clearance**

Low extraction drugs have minimal first-pass extraction and have almost complete hepatic bioavailability (e.g. warfarin, phenytoin)

PRACTICE QUESTION 6

6. The hepatic clearance and f_u for warfarin are reported as 4.60 mL/min and 0.03 respectively. What is the intrinsic clearance of warfarin? Assume hepatic blood flow is 1500 mL/min.

Givens:

$$CL_H = 4.60 \text{ mL/min}$$

$$f_u = 0.03$$

$$Q_H = 1500 \text{ mL/min}$$

Equation: $CL_h = \frac{Q \cdot f_u \cdot Cl_{int}}{Q + f_u \cdot Cl_{int}}$

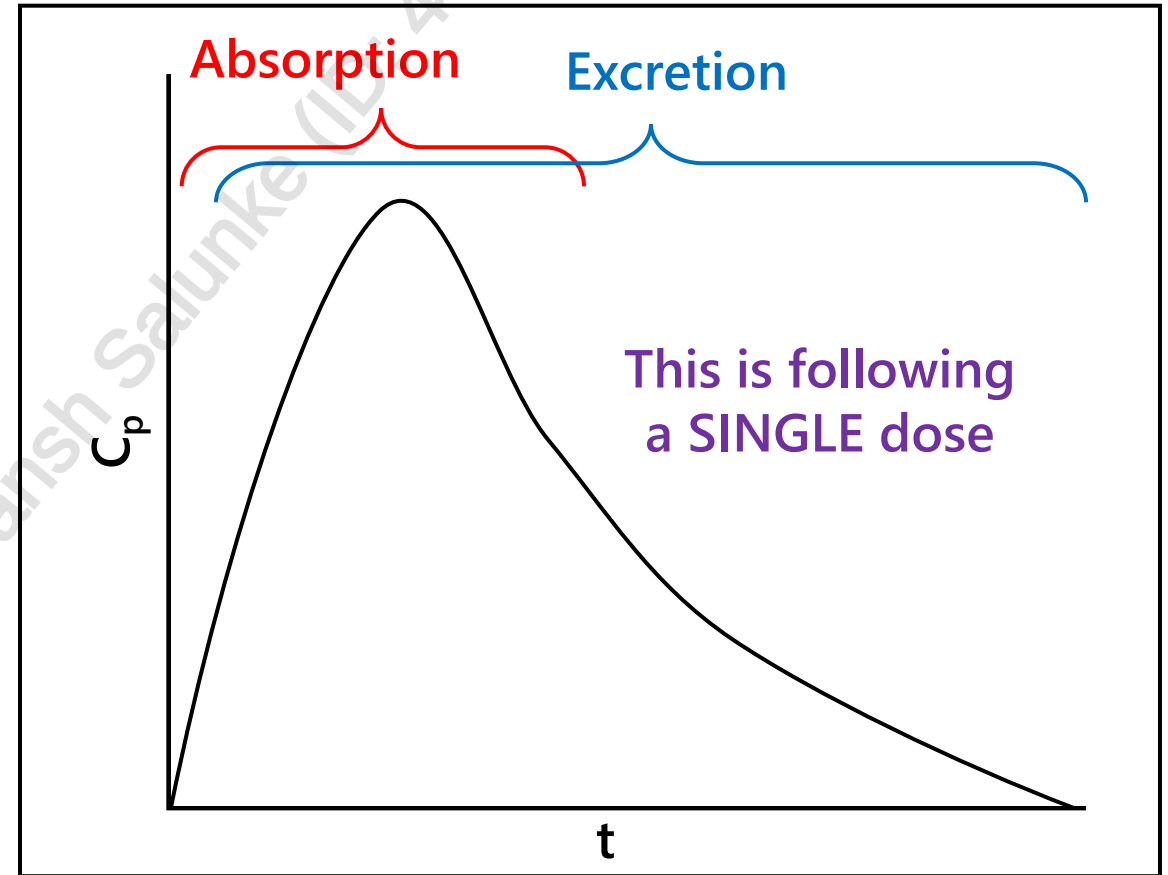
PRACTICE QUESTION 6- ANSWER

6. The hepatic clearance and f_u for warfarin are reported as 4.60 mL/min and 0.03 respectively. What is the intrinsic clearance of warfarin? Assume hepatic blood flow is 1500 mL/min.

$$\begin{aligned}Cl_h &= \frac{Q \cdot f_u \cdot Cl_{int}}{Q + f_u \cdot Cl_{int}} \\4.6 &= \frac{1500 \cdot 0.03 \cdot Cl_{int}}{1500 + 0.03 \cdot Cl_{int}} \\4.6 \cdot (1500 + 0.03 Cl_{int}) &= 45 Cl_{int} \\6900 + 0.138 Cl_{int} &= 45 Cl_{int} \\6900 &= 45 Cl_{int} - 0.138 Cl_{int} \\\frac{6900}{44.862} &= Cl_{int} \\Cl_{int} &= 153.8 \frac{mL}{min}\end{aligned}$$

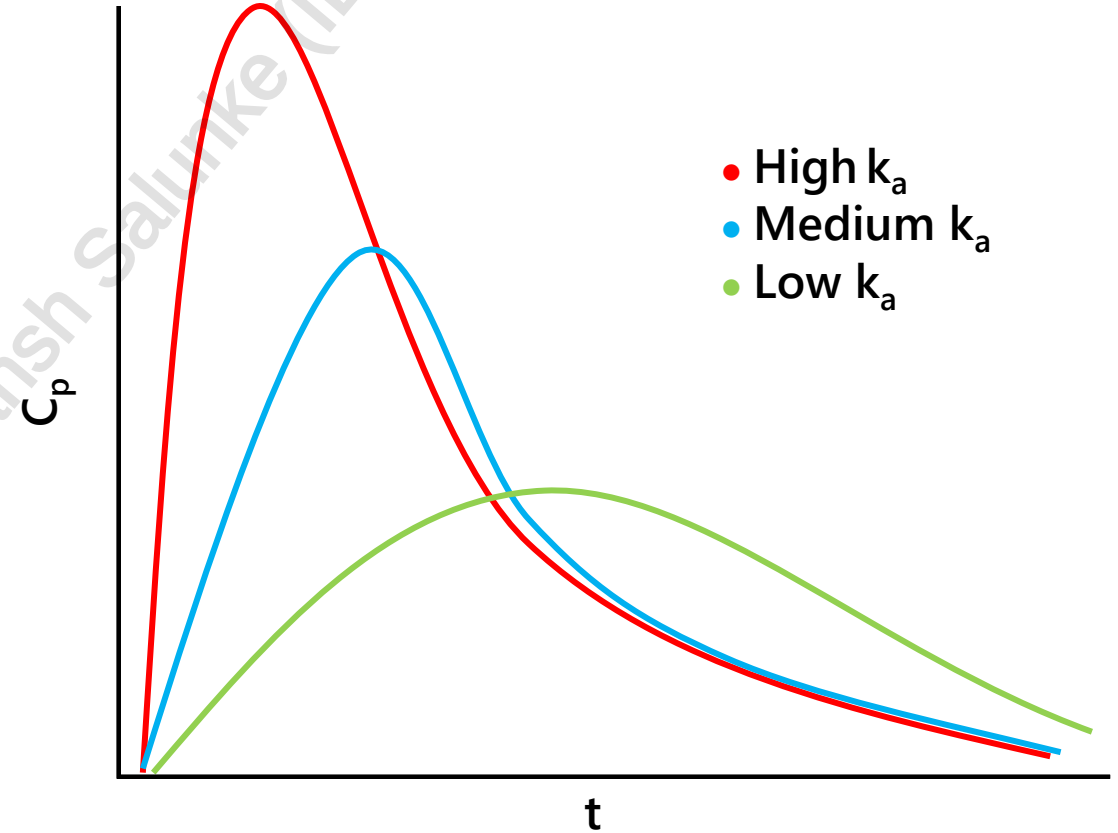
KINETICS OF ORAL DRUGS

- The concentration of drug in the system is a function of time
- Depends on dose, bioavailability, volume of distribution, absorption rates and elimination rates
- Initially absorption dominates, but then elimination is greater
- Modelled by a bi-exponential equation



ABSORPTION RATE CONSTANT (k_a)

- Rate of drug absorption according to first-order kinetics
- The HIGHER the k_a , the *faster* and *higher* the peak concentration



VOLUME OF DISTRIBUTION

- A theoretical volume relating the amount of drug in the body and the concentration of drug measured in the plasma (units: mL, L, L/kg)

$$V_d = \frac{\text{Amount of Drug in Body } (\sim \text{Dose})}{\text{Plasma Concentration } (C_p)}$$

- Does not refer to actual physiological volume: $V_d > 3\text{L}$ (plasma volume) suggests that the drug is distributed to other compartments (e.g. fat, liver, etc.)

Typical drug categories:

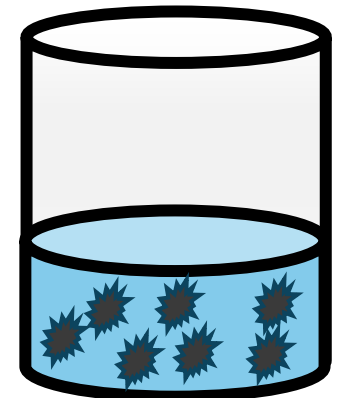
- 5 L (low; biologics) | 50 L (moderate; water sol' drugs) | 500 L (high; fat sol' drugs)

Example: change in volume of distribution with the addition of activated charcoal 🌟

Dose = 10 mg
 $C_p = 1 \text{ mg/L}$
 $V_d = 10 \text{ L}$



Dose = 10 mg
 $C_p = 0.1 \text{ mg/L}$
 $V_d = 100 \text{ L}$



CLEARANCE

A theoretical volume of plasma that is completely cleared of a drug in a given period of time (units: mL/min, L/h, $L \cdot h^{-1} kg^{-1}$)

- Proportional to the rate of drug elimination divided by the plasma drug concentration

$$CL = \frac{\text{Rate of Elimination (E)}}{\text{Plasma Concentration (C}_p\text{)}}$$

$$CL = \frac{\text{Dose (D)}}{AUC} \quad (\text{IV administration})$$

$$CL = \frac{\text{Bioavailability (F)} \cdot \text{Dose (D)}}{AUC} \quad (\text{non-IV administration})$$

Typical drug categories:

- < 1 mL/min (low; biologics) | 100 mL/min (moderate; fat sol' drugs) | > 1,000 mL/min (high; water sol' drugs)

ELIMINATION RATE CONSTANT

Fraction of drug amount in the body moved or eliminated per unit time (units: min⁻¹, h⁻¹)

- Independent of concentration
- Represented graphically as the slope of concentration versus time curve

$$k_{el} = \frac{\text{Clearance (CL)}}{\text{Volume of Distribution (V}_d\text{)}}$$

$$k_{el} = \frac{\ln(C_{p1}) - \ln(C_{p2})}{t_2 - t_1} = \frac{\ln(C_{p1}/C_{p2})}{\tau}$$

HALF LIFE

Duration of time required for the amount of drug to decrease by 50% (units: min, h)

- Based on first-order kinetics (concentration independent)
- Reflects two independent parameters: Volume of Distribution (Vd) and Clearance (Cl)
 - > 95% of a drug is eliminated after 5 half-lives

$$t_{1/2} = \frac{0.693}{k_{el}} = \frac{0.693 \cdot V_d}{Cl}$$

# of half-lives	Fraction of drug eliminated
1	0.5
2	0.75
3	0.875
4	0.9375
5	0.96875

PHARMACOKINETIC PARAMETERS

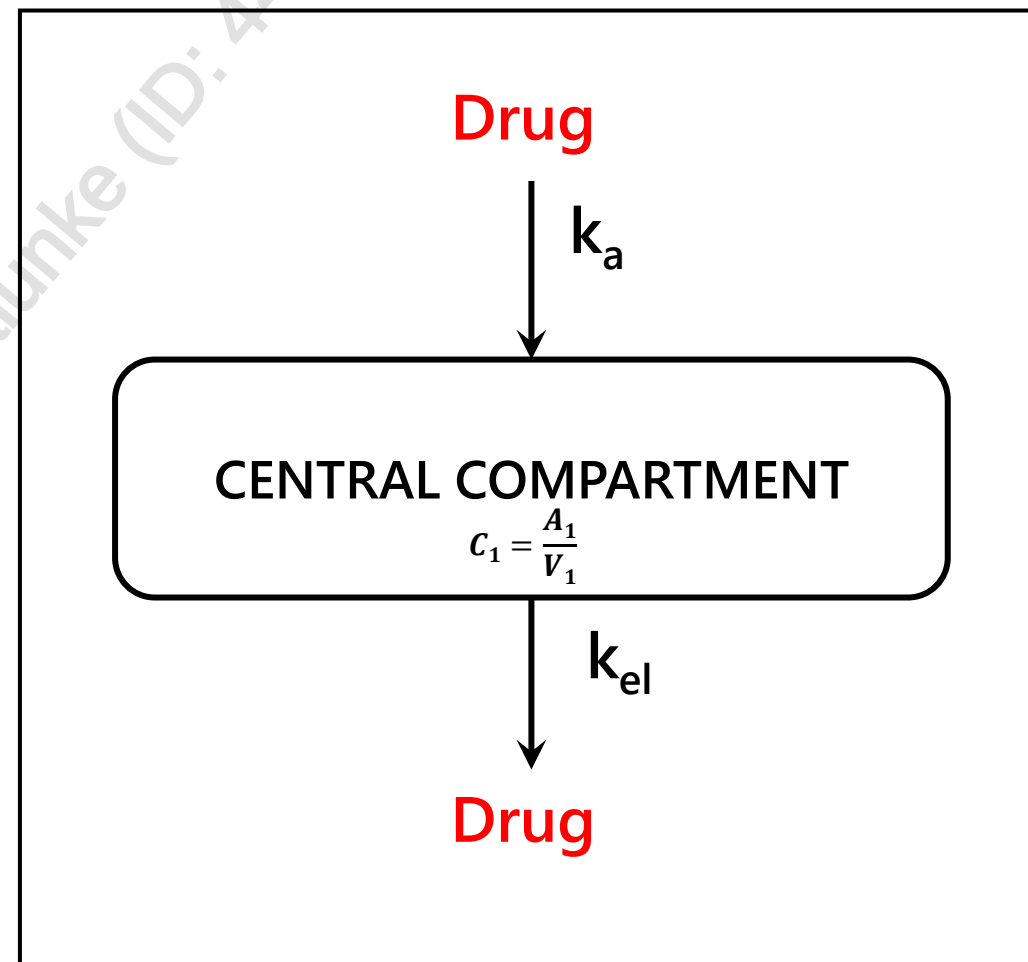
Symbol	Common Units	Definition
$C_{\max}$	ng/mL, $\mu\text{g/L}$, μM	Maximal [plasma] concentration
$T_{\max}$	h	Time to $C_{\max}$
AUC	ng $\cdot$ h/mL, ng $\cdot$ h ⁻¹ $\cdot$ mL ⁻¹ , $\mu\text{M} \cdot$ h	Area under the curve; exposure of [plasma] to drug
F	Unitless fraction or %	Bioavailability; fraction of the administered dose that reaches the systemic circulation
CL	mL/min, L/h, L $\cdot$ h ⁻¹ kg ⁻¹	Clearance; volume of [plasma] that is completely cleared of a drug in a given period of time
V_d	mL, L	Volume of distribution; theoretical volume relating the amount of drug in the body and the [plasma] concentration of drug
k_{el}	h ⁻¹	Elimination rate constant; proportionality coefficient of removal of drug from the body
k_a	h ⁻¹	Absorption rate constant; proportionality coefficient of entrance of drug into the body
$t_{1/2}$	min, h, days	Half-life; duration of time required for the amount of drug to decrease by 50%

RELEVANT QUESTIONS FOR PRACTICE

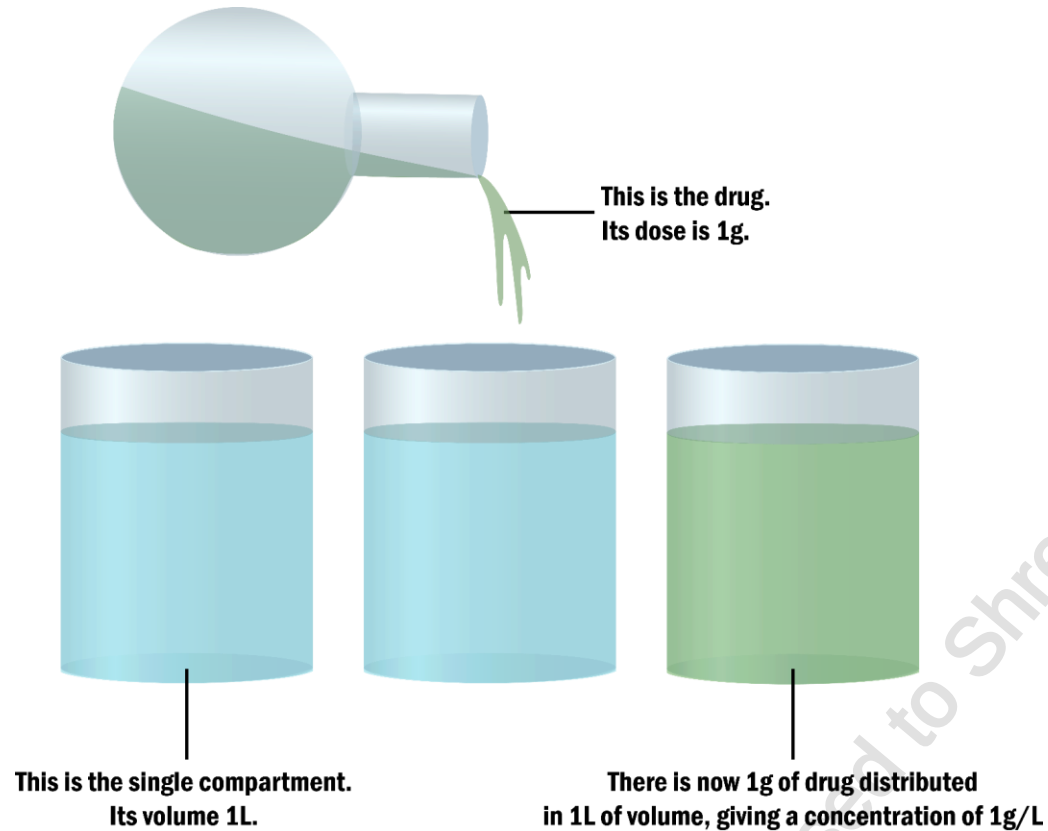
Question Bank Quiz	Questions
Pharmacokinetics Quiz #1	1, 2, 6, 7
Pharmacokinetics Quiz #2	All
Pharmacokinetics Quiz #3	All
Pharmacokinetics Quiz #4	1, 2, 10
Pharmacokinetics Quiz #5	6, 8-10
Pharmacokinetics Quiz #6	6-8
Pharmacokinetics Quiz #7	All
Pharmacokinetics Quiz #8	All
Pharmacokinetics Quiz #10	3-9

ONE-COMPARTMENT MODEL

- One-compartment model views the entire body as a **single central compartment**
- Drug undergoes rapid uptake into all tissues it can enter - "Well-Stirred"
- Drug distribution is **instantaneous** in this model
- Compartment is characterized by values
 - V_1 is the volume and is equal to the drug's volume of distribution
 - A_1 is the amount of drug in the compartment and is equal to the amount of drug in the body
 - C_1 is the concentration and is equal to the drug plasma concentration
 - k_a = rate of absorption
 - k_{el} = rate of elimination



ONE COMPARTMENT MODEL



ONE-COMPARTMENT PLASMA CONCENTRATION-TIME CURVE

One Compartment Model

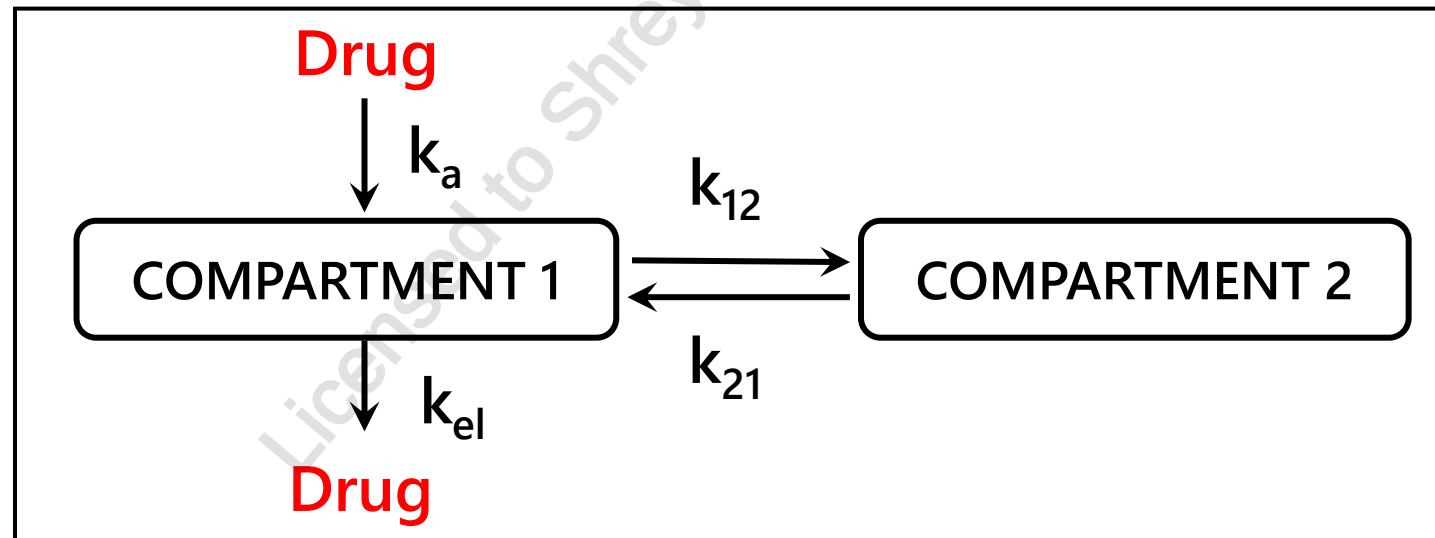
$\text{Log}(C_p)$

t

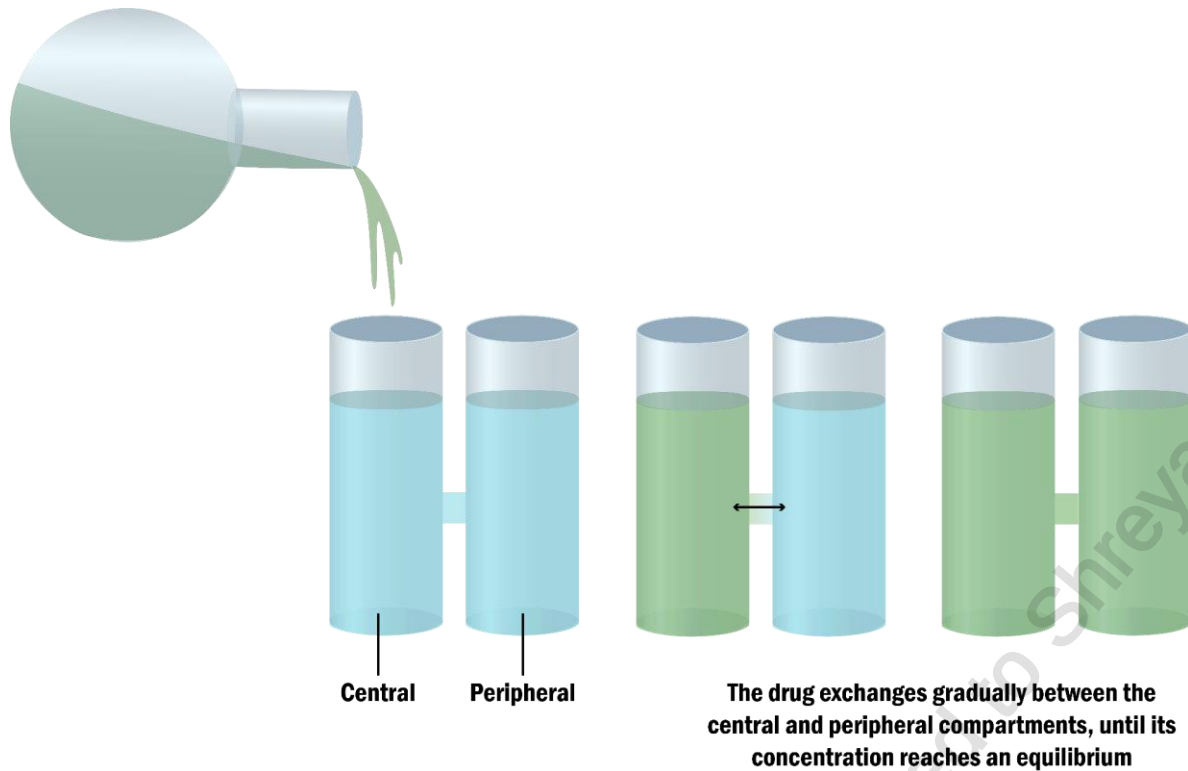
$$C_t = C_0 \cdot e^{-k_{el} \cdot t}$$
$$\ln(C_t) = \ln(C_0) - k_{el} \cdot t$$

TWO-COMPARTMENT MODEL

- Two-compartment model consists of the **central** compartment and a **peripheral** compartment
- Peripheral compartment consists of tissue where uptake is slower than central compartment
- Volume, drug amount, and concentration symbols for peripheral compartment denoted with a 2 (V_2 , A_2 , C_2)
- Distribution to peripheral compartment driven by A_1 , with rate constant k_{12}
- Redistribution to the central compartment is driven by A_2 , with rate constant k_{21}
- Rate constant for elimination is k_{10}

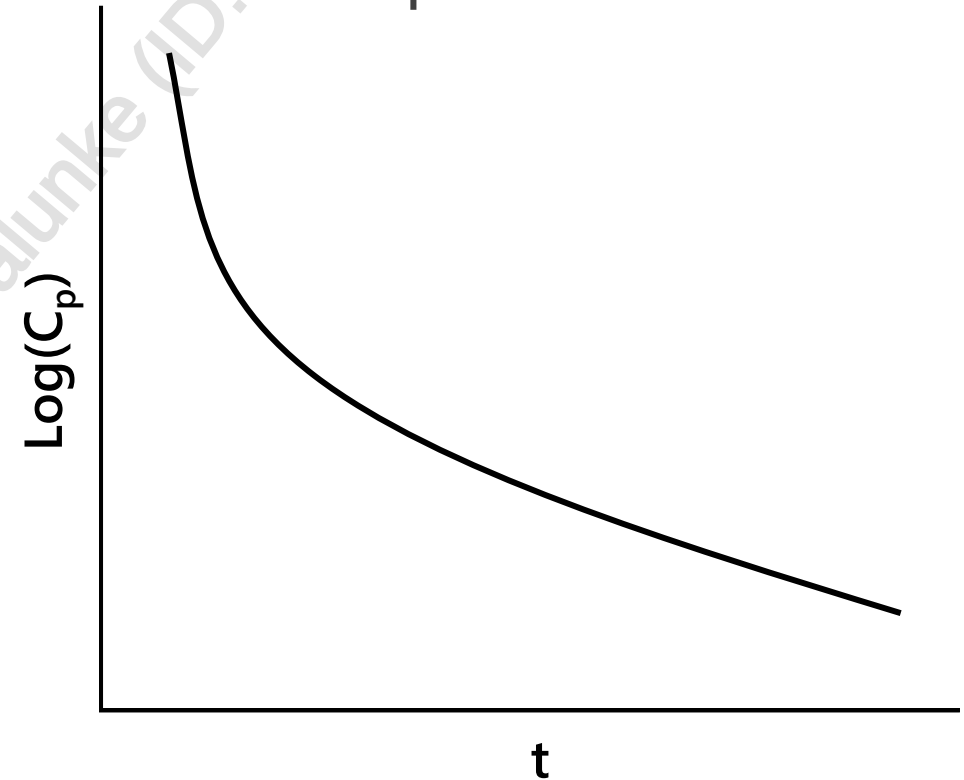


TWO COMPARTMENT MODEL



TWO-COMPARTMENT PLASMA CONCENTRATION-TIME CURVE

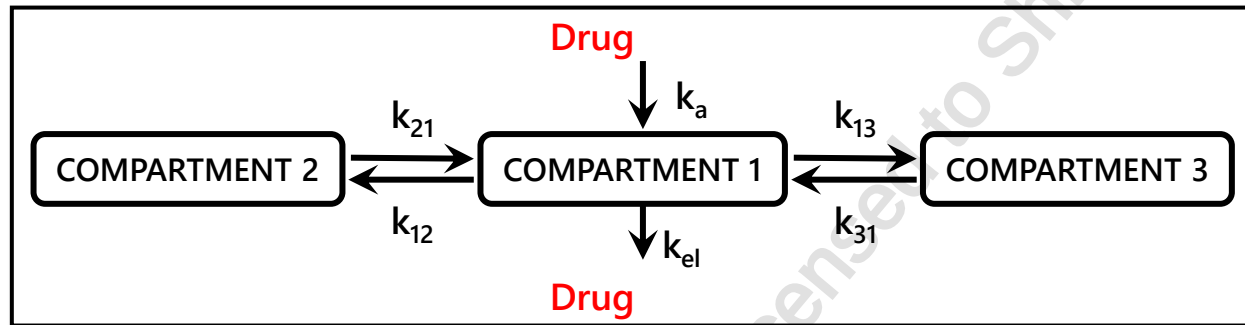
Two Compartment Model



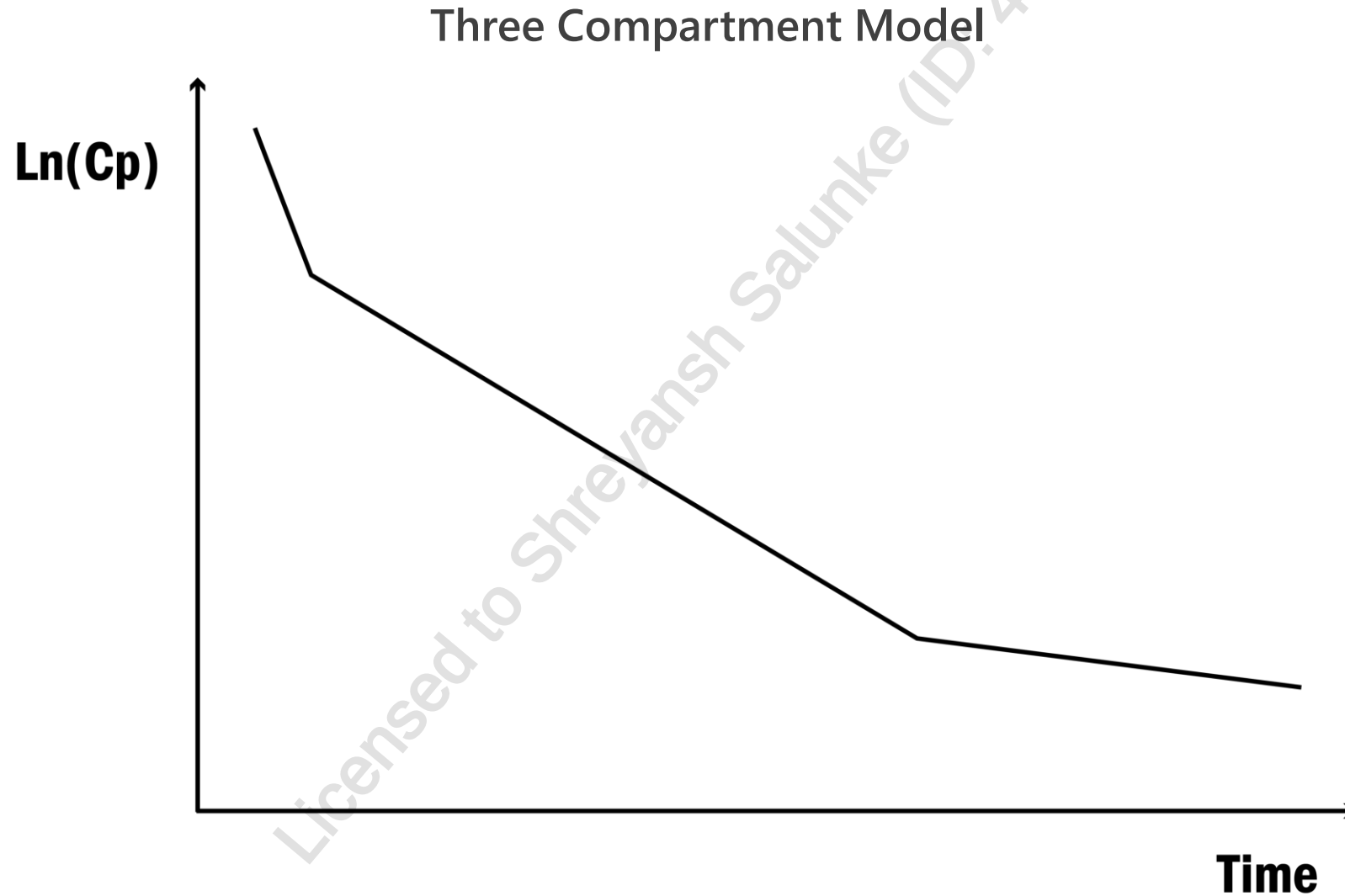
$$C_t = C_1 \cdot e^{-\alpha \cdot t} + C_2 \cdot e^{-\beta \cdot t}$$

THREE-COMPARTMENT MODEL

- Three-compartment model consists of the **central** compartment, **peripheral** compartment, and **deep tissue** compartment
- Deep tissue compartment consists of **poorly perfused tissue** that the drug distributes into very slowly
- Deep tissue volume, drug amount, and concentration denoted with a 3 (V_3 , A_3 , C_3)
- Distribution constant is k_{13} ; redistribution constant is k_{31}
- Rate of redistribution from deep tissue compartment **controls final stages of elimination**



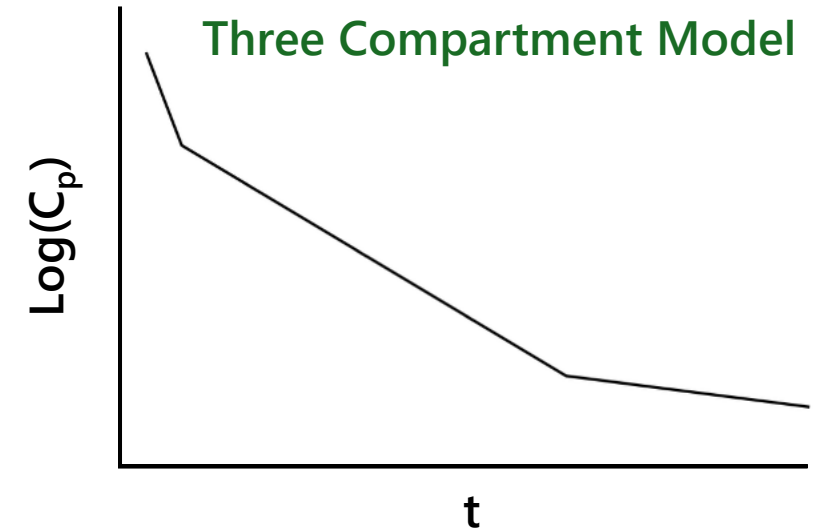
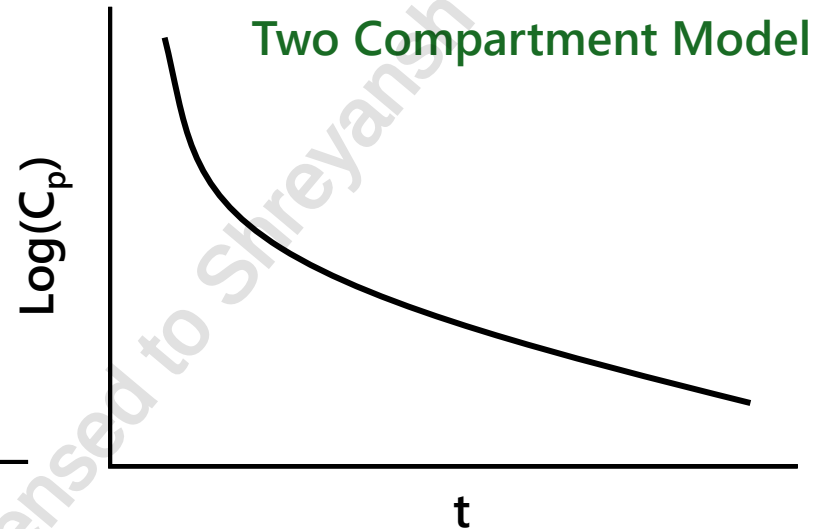
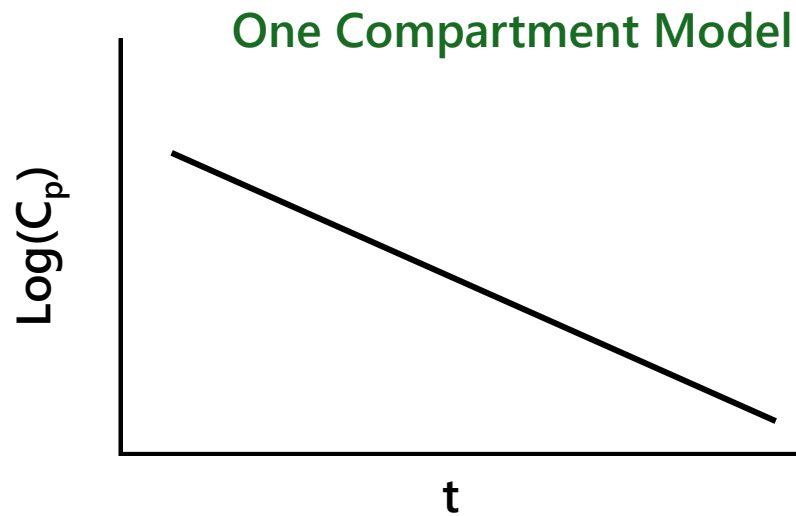
THREE-COMPARTMENT PLASMA CONCENTRATION TIME CURVE



DRUG CONCENTRATIONS VERSUS TIME CURVES

DETERMINING WHICH MODEL THE DRUG FOLLOWS

The number of slopes [phases] visible on a semi-log plot can be used to estimate the number of compartments a model should have



INTRAVENOUS INJECTION

- A route of administration that **directly** puts drug into systemic circulation, providing **immediate** therapeutic plasma concentration (**recall $F = 100\%$!**)

Intravenous route of administration

Bolus

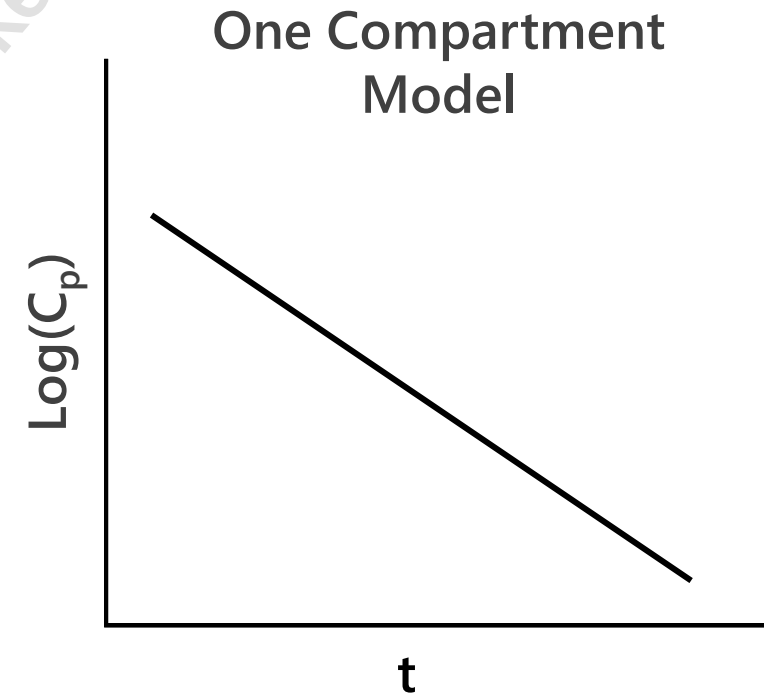
- **Single** dose administration of medication in a short period of time

Infusion

- Dose that is administered over an **extended** period
- Can be **intermittent** (e.g. at regular intervals) or **continuous** (e.g. at constant rate)

ONE COMPARTMENT IV ADMINISTRATION

- Plasma concentration influenced by first-order elimination only
- Plasma-time profile presents a straight line on a semilogarithmic scale
- Initial amount of drug at time zero is the effective dose
- Drug distribution from plasma to tissue instantaneous
- Concentration in compartment equal to plasma concentration
- Volume of compartment is equal to volume of distribution ($V_1 = V_d$)



IV BOLUS PLASMA CONCENTRATION

- The rate of drug over time at the start is equal to the difference between the rate of inputs and the rate of outputs

The plasma concentration is found by integrating the rate of drug over time from zero to infinity:

$$C_p = \frac{F \cdot D}{V_d} \cdot e^{-kt} = C_{p_0} \cdot e^{-kt}$$

- C_p is the plasma concentration
- V_d is the volume of distribution
- F is the bioavailability factor (=1 since IV)
- D is the dose administered
- C_{p_0} is the plasma concentration at time zero
- k is the elimination rate constant

SAMPLE PROBLEM

7. A 50-mg dose of a drug is administered via an intravenous injection. The drug has a volume of distribution of 100 L, and an elimination rate constant of 0.2 h^{-1} ?
- a) What is the initial plasma concentration?
 - b) What is the plasma concentration 4 hours after administration?

PRACTICE QUESTION 7

7. A 50-mg dose of a drug is administered via an intravenous injection. The drug has a volume of distribution of 100 L, and an elimination rate constant of 0.2 h⁻¹?
- a) What is the initial plasma concentration?

$$Cp_0 = \frac{F \cdot D}{Vd}$$

$$Cp_0 = \frac{1 \cdot 50 \text{ mg}}{100 \text{ L}}$$

$$Cp_0 = 0.5 \frac{\text{mg}}{\text{L}}$$

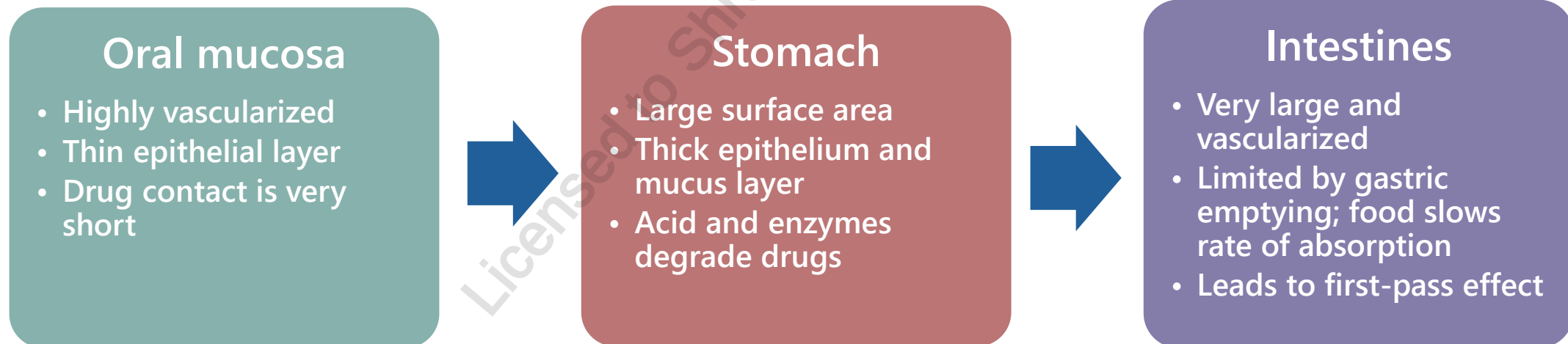
PRACTICE QUESTION 7- ANSWER

7. A 50-mg dose of a drug is administered via an intravenous injection. The drug has a volume of distribution of 100 L, and an elimination rate constant of 0.2 h⁻¹?
- b) What is the plasma concentration 4 hours after administration?

$$C_p = \frac{F \cdot D}{V_d} \cdot e^{-kt}$$
$$C_p = \frac{1 \cdot 50 \text{ mg}}{100 \text{ L}} \cdot e^{-0.2 \text{ h}^{-1} \cdot 4 \text{ h}}$$
$$C_p = 0.22 \frac{\text{mg}}{\text{L}}$$

ORAL ADMINISTRATION

- Most common form of administration
- Sites of absorption: oral mucosa, stomach and intestines
- Obstacles: stomach acid, proteases, poor diffusion, transit time
- Physiological considerations:
 - Absorption highly enhanced in thin, vascularized tissue
 - Acids can ionize many drugs, but can unionize drugs with a low pK_a
 - Food will often slow absorption, this may be beneficial for a sustained release
 - All drugs absorbed in the intestines are subject to the first-pass effect



PLASMA CONCENTRATION AFTER ORAL DOSE

The plasma concentration at any time after a dose can be found using the following biexponential equation:

$$C_p = \frac{F \cdot D \cdot k_a}{V_d \cdot (k_a - k)} \cdot (e^{-kt} - e^{-k_a t})$$

- C_p is the plasma concentration
- V_d is the volume of distribution
- k is the elimination rate constant
- F is the bioavailability
- D is the dose
- k_a is the absorption rate constant
- t is time

MODIFIED RELEASE PRODUCTS

- Immediate release of a drug is not always viable or desired; drug release rate and/or timing can be controlled
- Allows for greater compliance (e.g. decreased frequency of administration)
- Can help targeting and protect drugs; increases utility and range of oral drugs

Extended-release drugs

- Drugs are released over an extended period of time
- Useful with drugs that are cleared quickly and have targeted bioavailability
- Repeat action: releasing a dose at constant intervals

Delayed release drugs

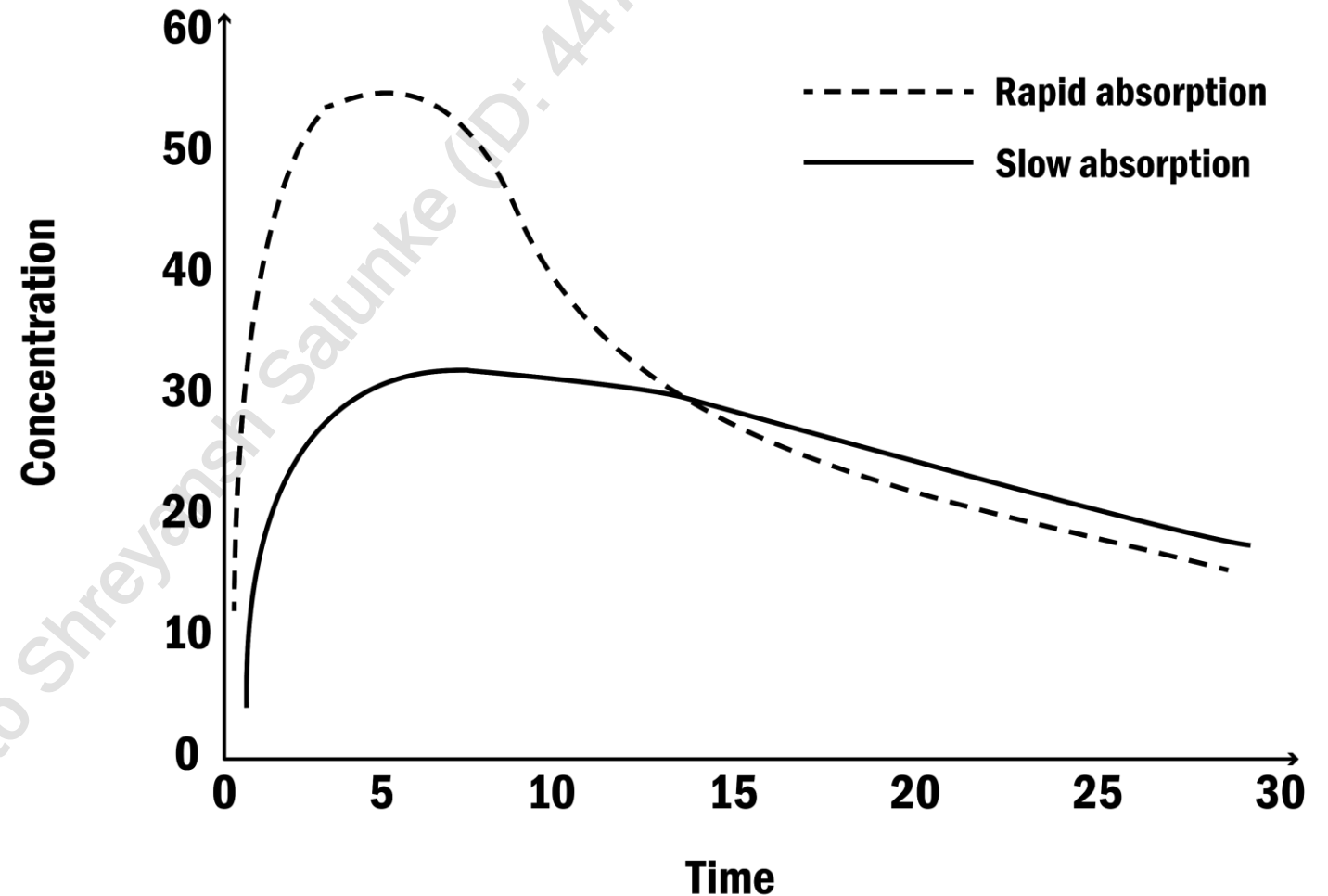
- Coatings prevent immediate release of drugs; often used to prevent drug-stomach interactions (e.g. acetylsalicylic acid)

Target release drugs

- Drugs packed in carriers such as liposomes, micelles, etc.
- Carriers are absorbed and transported to targets
- Allows for precise targeting of harmful drugs

FLIP-FLOP KINETICS

- The terminal fall in plasma concentration is controlled by k_a
- This is the flip-flop model
- Most prolonged-release preparations follow the flip-flop model
- Flip-flop kinetics occur when **the absorption rate is much slower than the elimination rate**



NEXT CLASS: PHARMACOKINETICS PART II

- Multiple Doses and Steady-State
- Therapeutic Drug Monitoring
- Non-linear PK
- Interactions
- Pharmacodynamics

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CHANGE LOG

July 2025

- Slide 13: added sentence under pH-partition hypothesis
- Slide 14-15: fixed Noyes-Whitney equation
- Slide 72: added 3-compartment graph for completeness
- Minor formatting changes made throughout

October 2025

- Slide 13: added examples of drug salts
- Slide 19: added antilog definition for clarity
- Slide 75 : removed F from equation since IV bolus ($F=1$)
- Minor formatting changes made throughout

February 2026

- No content changes made